

BRIDGE: BALANCING REPRESENTATIONS BY IDENTIFYING DEFICITS AND GENERATING EXAMPLES

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ABSTRACT

Self-supervised learning inherits the coverage of its source data. When an unlabeled corpus is long-tailed, the learned representation becomes dense around frequent modes and thin where coverage is weak, and a change of training objective adds none of the missing support. We introduce BRIDGE, Balancing Representations by Identifying Deficits and Generating Examples, a label-free feedback loop in which the current encoder scores every source image by its mean k NN distance, a budgeted acquisition rule selects a sparse but recoverable band of that score distribution, a frozen image-to-image diffusion model adds local variants of the selected anchors, and pretraining continues on the enlarged source set. We formulate acquisition as recoverability-weighted allocation of local support and derive the conditions under which a finite budget should favor sparse interior regions and be reallocated after every repair. Across five long-tailed and web-sourced regimes, BRIDGE gives the best seven-task linear-probe and k NN transfer, and paired experiments extend the gain to ViT-S under contrastive, self-distillation and masked-reconstruction objectives, to low-label fine-tuning and to semantic segmentation. Generated repair matches an oracle that restores the real images withheld when the long tail was built, and under a frozen encoder the inserted images shrink the neighborhoods of the selected anchors by twice the amount observed at controls matched on initial sparsity.

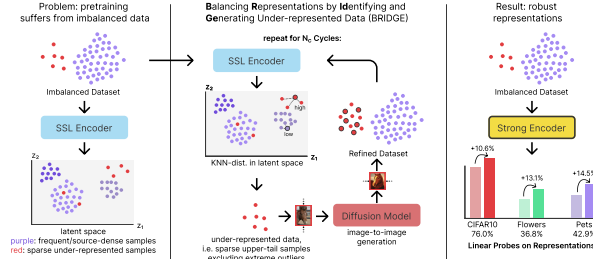


Figure 1: Overview of BRIDGE. Pretraining, sparsity scoring in the current representation and targeted image-to-image generation alternate over cycles. The budget is spent on a mode-centered band inside the upper tail of the score distribution, which excludes the extreme tail.

1 INTRODUCTION

Self-supervised learning is the standard route to transferable visual representations from unlabeled images (Chen et al., 2020; He et al., 2020; Caron et al., 2021; He et al., 2022; Oquab et al., 2024; Siméoni et al., 2025), and every objective learns from whatever the source corpus contains. When that corpus is long-tailed, the representation concentrates around frequent modes, covers rare regions weakly and transfers worse (Tian et al., 2021; Assran et al., 2023; Kukleva et al., 2023; Jiang et al., 2021). Prior work has responded on the model side, with a temperature schedule (Kukleva et al., 2023), a pruned self-competitor (Jiang et al., 2021) or imported external data (Bai et al., 2023), and all of it leaves the source set fixed.

We ask whether the source set can be repaired while learning proceeds. BRIDGE, Balancing Representations by Identifying Deficits and Generating Examples, embeds the current source set

with the current encoder after each pretraining stage, scores every image by its mean distance to its k nearest neighbors, selects a sparse but semantically coherent band of that distribution, and generates local variants of the selected images with a frozen image-to-image diffusion model, after which pretraining continues on the enlarged set (Figure 1). The encoder identifies where its own representation is thin and the generator adds support there. No labels, class frequencies or downstream signals enter the loop, and Section 6.5 verifies the mechanism directly, since under a frozen encoder the generated images shrink the selected anchors’ neighborhoods by twice the amount observed at matched untreated images.

Two decisions determine whether such a loop helps, where a finite generation budget is spent and whether the allocation is re-estimated after each repair. BRIDGE restricts acquisition to the 75th to 99th percentile band of the score distribution, centers a candidate pool on the mode of that band and covers the pool with farthest-point sampling. We formalize both decisions as budgeted allocation over regions of representation space with diminishing returns to local support and repairability that falls in the extreme tail. The optimal band is then interior, greedy reallocation after each intervention is optimal, and any rule applied once has an effect bounded by the ratio of budget to existing support. The experiments confirm the last statement, since nine score bands from the densest quartile to the extreme tail are indistinguishable after one repair stage and the same rules separate by 8.0 points over five.

- We formulate label-free data repair as recoverability-weighted allocation of local support and prove that sparse interior regions carry the largest marginal value, that greedy reallocation after each repair is optimal under diminishing returns, and that any single-stage allocation has an effect bounded by the ratio of budget to existing support.
- We instantiate this policy in BRIDGE, where mean k NN distance estimates local support, a trimmed mode window selects a recoverable sparse band, farthest-point sampling limits redundancy and a frozen image-to-image generator supplies variants.
- We evaluate five source regimes and seven transfer tasks under linear-probe and k NN evaluation, with paired ViT-S experiments across four SSL objectives, low-label fine-tuning, semantic segmentation, an oracle that restores withheld real images, and matched controls for the closest acquisition rules, repair operators and generator priors.
- We separate the acquisition rule from the loop by running identical rules under one repair stage and under five, verify where the generated support lands under a frozen encoder, and locate the effect in the repeated re-estimation.

2 RELATED WORK

Self-supervised learning under skewed sources. The sensitivity of contrastive, self-distillation and masked-reconstruction pretraining (Chen et al., 2020; He et al., 2020; Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025; He et al., 2022) to source imbalance is established. Divide and Contrast clusters uncurated data into specialized experts (Tian et al., 2021), the hidden uniform cluster prior explains why standard objectives favor balanced sources (Assran et al., 2023), and FASSL reweights by estimated class frequency (Lin et al., 2023). The methods closest to our comparison change the objective on a fixed corpus, SDCLR through a pruned self-competitor that gives persistently difficult samples more gradient (Jiang et al., 2021), Temperature Schedules by emphasizing head and tail at different times (Kukleva et al., 2023), and COLT by drawing external out-of-distribution images into the batch (Bai et al., 2023). BRIDGE holds the objective fixed and changes the source set, so it is complementary, and we report the combination with TS and SDCLR.

Density, rarity and representation-driven curation. Mean deep nearest-neighbor distance is an effective nonparametric out-of-distribution score in high-dimensional feature spaces (Sun et al., 2022), and we use it as a local-support estimate on the training set itself. Representation-driven curation prunes by prototype distance (Sorscher et al., 2022) or concept-cluster complexity (Abbas et al., 2024) and treats data quality as a subject in its own right (Whang et al., 2023), and all of it removes examples. SMOTE adds them by interpolating labeled minority examples (Chawla et al., 2002), and DOPING oversamples the low-density edge of a learned latent distribution for unsupervised anomaly detection (Lim et al., 2018). DOPING is the closest antecedent to acquiring by latent rarity, targets

the extreme edge, uses no image generator and never recomputes its target, and our top-tail control is a DOPING-style rule under our generator and budget.

Budgeted acquisition and coverage. Active learning is the formal precedent for spending a finite acquisition budget under a learned representation, and the core-set formulation treats acquisition as k -center coverage and selects with the greedy farthest-first traversal we use inside the mode window, which carries a factor-two approximation guarantee (Sener and Savarese, 2018; Gonzalez, 1985), while ProbCover maximizes covered probability mass at a fixed radius (Yehuda et al., 2022). Hachohen et al. (2022) show that the preferred strategy reverses with budget, dense regions being better targets when the budget is small relative to the corpus, which our single-stage sweep reproduces. Active learning acquires labels for existing images, whereas BRIDGE creates images without labels and its scoring function is the encoder being trained.

Generative augmentation for supervised recognition. SDEdit adds noise to an image and denoises it under a diffusion prior (Ho et al., 2020; Rombach et al., 2022; Meng et al., 2022), and that operator is our repair step. DA-Fusion applies it to few-shot supervised recognition with class prompts and domain adaptation (Trabucco et al., 2024). Class-conditional generation improves supervised ImageNet accuracy (Azizi et al., 2023), with a utility that depends on prompting, filtering, diversity and domain gap (He et al., 2023; Fan et al., 2024). TADA, the closest targeted method, identifies examples a supervised classifier learns slowly, generates faithful variants for them, and reports that augmenting a targeted third of the data beats augmenting all of it (Nguyen et al., 2026). Every method in this family uses class identity or a supervised loss to decide what to augment and decides once. BRIDGE uses neither and decides again after every repair, and we adapt the TADA criterion to a label-free learning-speed signal as a direct baseline.

Synthetic data for representation learning. Synthetic imagery has served as a training source for self-supervised features since Ren and Lee (2018), from procedural noise and image programs (Baradad et al., 2021; 2022), latent neighborhoods of a black-box generator (Jahani et al., 2022), synthetic ImageNet clones (Sariyildiz et al., 2023) and text-to-image samples used as multi-positive views (Tian et al., 2023). These methods replace or globally supplement the source and allocate no budget over an observed skewed corpus. DiffAug, the closest iterative predecessor, trains a semantic encoder and a conditional diffusion generator jointly so that generated positives improve contrastive learning (Zang et al., 2024). Its generator must be learned from the same small skewed source it is meant to repair, whereas BRIDGE holds a pretrained generator fixed and asks where it should intervene, and our DiffAug control adapts its objective to our source.

Data engines and external priors. Industrial data engines close a loop between a deployed model and its training data. AIDE uses vision-language and language models to diagnose detector failures, curate and auto-label data and verify the result (Liang et al., 2024), and Omniverse Replicator supplies infrastructure for iterating such pipelines (Worker and Pathak, 2022). Both operate from supervised task failures with annotation in the loop and acquire by the occupancy of semantic clusters, which our inverse cluster-frequency control reproduces under the same generator. A frozen generator also carries knowledge from its own pretraining data, which DreamTeacher distills into discriminative backbones (Li et al., 2023), DIFT reads out as semantic correspondence (Tang et al., 2023) and REPA transfers in the other direction (Yu et al., 2025), so distilling the generator’s features directly is a necessary control, which we report. Appendix A tabulates the closest methods by supervision, acquisition signal, data action and feedback.

3 THEORETICAL FRAMEWORK

We model targeted repair as budgeted allocation over coherent regions of representation space. A repair is useful only when it adds support where downstream risk is concentrated and the generator preserves the semantic content of its anchor, so the policy must weigh support deficit against reparability and redundancy. Let $r_k(z; Z)$ be the distance from z to its k th nearest neighbor in Z and let A be a finite set of valid variants generated from z . Proofs are in Appendix C.

Proposition 1 (Monotone local support). *For any finite A , $r_k(z; Z \cup A) \leq r_k(z; Z)$. The inequality is strict when an added point enters the first k neighbors of z and displaces a neighbor at strictly larger distance.*

To connect support to learning, partition the representation into regions j with current effective support n_j and model the regional contribution to downstream error by

$$\mathcal{B}(\mathbf{m}) = \sum_j p_j a_j (n_j + m_j + \kappa)^{-\gamma}, \quad (1)$$

where p_j is the transfer-relevant mass of region j , a_j its difficulty, m_j the number of valid repairs allocated to it, and $\kappa, \gamma > 0$, the usual diminishing-returns form for learning from additional samples. Write $\Delta_j(m_j) = p_j a_j [(n_j + m_j + \kappa)^{-\gamma} - (n_j + m_j + 1 + \kappa)^{-\gamma}]$ for the reduction from one more valid repair.

Proposition 2 (Sparse-region marginal value). *$\Delta_j(m_j)$ is positive and strictly decreasing in $n_j + m_j$. Between two regions of equal transfer mass and difficulty, the region with less effective support has the larger marginal value.*

Sparsity alone is not a sufficient target, because generation fails on corrupt or incoherent anchors. Let d be a sparsity score, $b(d)$ the marginal value of a valid repair at that score, $\rho(d)$ the probability of a faithful variant, and $\lambda \geq 0$ the cost of an unfaithful one. Expected repair utility is

$$u(d) = \rho(d) b(d) - (1 - \rho(d)) \lambda. \quad (2)$$

Proposition 3 (Interior recoverable-support optimum). *Suppose b and ρ are positive and differentiable on $[d_{\min}, d_{\max}]$ with $b'(d) > 0$ and $\rho'(d) < 0$, and define $g(d) = b'(d)/(b(d) + \lambda) + \rho'(d)/\rho(d)$. If g is strictly decreasing with $g(d_{\min}) > 0$ and $g(d_{\max}) < 0$, then u has a unique maximizer d^* in the interior, characterized by $g(d^*) = 0$.*

The condition compares the relative growth of support value against the relative decay of repairability, and since its monotonicity assumption on g is close to assuming the conclusion, Proposition 6 in the appendix discharges it for $b(d) = b_0 d^\beta$ with $\beta \in (0, 1]$ and $\rho(d) = \rho_0 e^{-\gamma d}$, where g is strictly decreasing for every $\lambda \geq 0$ and $d^* = \beta/\gamma$ at $\lambda = 0$. The optimum moves outward when support value is closer to linear and inward when fidelity decays faster. BRIDGE operationalizes the interior preference with a trimmed score band and estimates neither parameter. The remaining results concern repetition, so consider a separable objective $U(\mathbf{m}) = \sum_j U_j(n_j + m_j)$ with each U_j increasing and discretely concave and a budget $\sum_j m_j = B$.

Proposition 4 (Adaptive allocation under diminishing returns). *For integer allocations, repeatedly assigning one repair to the region with the largest current marginal gain maximizes $U(\mathbf{m})$. A fixed allocation becomes suboptimal once a region’s marginal gain falls below that of another region.*

Proposition 5 (Single-stage insensitivity). *Let $\mathbf{m}$ and $\mathbf{m}'$ be any two allocations of the same budget B under Equation 1, computed from one representation and applied without re-estimation. Then*

$$|\mathcal{B}(\mathbf{m}) - \mathcal{B}(\mathbf{m}')| \leq B \gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}, \quad (3)$$

which tends to zero as $B/n_{\min} \rightarrow 0$ for the least-supported region $n_{\min}$.

With 2,500 generated images against a source of about 10,000, Proposition 5 predicts that no acquisition rule separates from any other after a single stage, which Section 6.4 confirms. The five-stage separation lies outside this single-stage model, since re-estimation makes each score depend on earlier allocations, and we claim no matching lower bound for the repeated regime. Equation 1 and the monotone behavior of b and ρ are modeling assumptions, and Section 6 tests their observable predictions, an interior optimum, a large gap to uniform and extreme-tail acquisition, a null effect of the rule at one stage, and added support at the selected anchors.

4 BRIDGE: BALANCING REPRESENTATIONS BY IDENTIFYING DEFICITS AND GENERATING EXAMPLES

4.1 LOOP AND SCORING

Let $D^{(0)}$ be an unlabeled source set and f_θ an SSL encoder. BRIDGE alternates pretraining and repair over C cycles, and at cycle c the encoder is $f_{\theta^{(c)}}$, the source set $D^{(c)}$ and its embedding set $Z^{(c)}$. After each training stage, scoring embeds every current source image and assigns a label-free

sparsity score, selection turns that distribution into an anchor set under a generation budget B , and repair conditions the frozen generator on the anchors and appends the variants. The encoder is the only learned component, so the loop applies to any objective that yields embeddings. Every cycle continues from the previous encoder with a fresh optimizer schedule and retains all earlier images, so a cycle costs one pass over the grown source set and never a pretraining run from scratch.

After training on $D^{(c)}$ we embed each image, $z_i^{(c)} = f_{\theta^{(c)}}(x_i)$, normalize to unit length, and compute

$$d_i^{(c)} = \frac{1}{k} \sum_{z \in \mathcal{N}_k(z_i^{(c)}; Z^{(c)} \setminus \{z_i^{(c)}\})} \|z_i^{(c)} - z\|_2^2, \quad (4)$$

where $\mathcal{N}_k(z; Z)$ is the set of k nearest neighbors of z in Z , so a large score marks an image in a sparse part of the current representation. We use $k = 100$, and k between 10 and 200 moves the seven-task average by about one point (Appendix Table 20).

4.2 SPARSE AND RECOVERABLE ANCHOR SELECTION

The extreme tail is populated by artifacts, degenerate crops and images the generator cannot vary faithfully, and uniform sampling spends most of its budget in regions that are already dense. The selector targets a sparse region that is still internally coherent in three steps (Figure 2). It restricts attention to $U^{(c)} = \{i \mid q_{0.75}^{(c)} \leq d_i^{(c)} \leq q_{0.99}^{(c)}\}$ and builds a histogram over that band whose modal bin has center $m^{(c)}$. Given a candidate-pool multiplier α , it forms the mode-centered pool $C^{(c)} \subseteq U^{(c)}$ of size αB that minimizes $\sum_{i \in C} |d_i^{(c)} - m^{(c)}|$. It then selects the anchor set $S^{(c)}$ of size B by greedy farthest-point sampling in embedding space from that pool. The band excludes the scores where faithful repair is least reliable, the mode favors a region with repeated empirical support over isolated endpoints, and farthest-point sampling keeps the budget off near-duplicates. We use $\alpha = 4$ and $B = 500$.

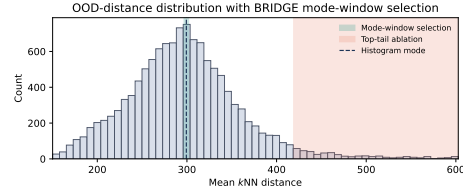


Figure 2: The score distribution of one ImageNet-100-LT run. The mode window centers the candidate pool on the modal bin of the 75th to 99th percentile band (green). The top-tail control spends the same budget in the shaded extreme tail.

4.3 LOCAL GENERATIVE REPAIR AND CONTINUED PRETRAINING

For each anchor we generate $G = 5$ variants with a frozen image-to-image diffusion model g_ϕ , conditioning on the anchor image under an empty text prompt so that the repair signal comes from the image and never from a class name, and set $D^{(c+1)} = D^{(c)} \cup \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$. The noise ϵ_j indexes independent translations of one anchor, so variants keep coarse structure while changing texture, pose, color or composition. The main configuration is Stable Diffusion 3 Medium with 20 denoising steps, strength 0.6 and output resolution equal to the training crop. Five training stages contain four repairs, so a full run adds 10,000 images to a source of about 10,000. The score estimates support deficit, the mode window enforces recoverability, farthest-point sampling approximates coverage and cyclic re-estimation reallocates the budget after every intervention, the four requirements of Section 3. Algorithm 1 in the appendix gives the loop.

5 EXPERIMENTAL SETUP

Sources, targets and protocol. We pretrain on five unlabeled source regimes, the long-tailed ImageNet-100-LT (Liu et al., 2019), CIFAR-10-LT and CIFAR-100-LT (Krizhevsky, 2009), and 10k subsets of the web corpora PASS (Asano et al., 2021) and DiffusionDB (Wang et al., 2023). Transfer is measured on seven benchmarks, CIFAR-10-LT, CIFAR-100-LT, Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), Oxford Flowers (Nilsback and Zisserman, 2008), Oxford-IIIT Pets (Parkhi et al., 2012) and ImageNet-100-LT, by linear probe and by k NN classification on frozen features. Every number is a three-seed mean and standard deviation, and paired comparisons share seeds and source checkpoints. ResNet-50 is the default backbone because it is common to the

Table 1: Linear-probe transfer for five source regimes, three label-derived long-tailed sources and two 10k web corpora, on seven downstream tasks, mean $\pm$ standard deviation over three seeds. Every method pretrains on the source in the left column under the same schedule. Hybrids of BRIDGE with TS and SDCLR are in Appendix Tables 9 and 10.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.8 $\pm$ 0.2	49.3 $\pm$ 0.2	16.4 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.5 $\pm$ 0.7
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	78.1 $\pm$ 0.3	49.8 $\pm$ 0.4	9.2 $\pm$ 0.8	17.7 $\pm$ 1.0	18.2 $\pm$ 1.1	27.4 $\pm$ 2.7	39.0 $\pm$ 0.6	34.2 $\pm$ 1.0
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	74.2 $\pm$ 0.5	48.1 $\pm$ 0.4	8.3 $\pm$ 1.0	18.2 $\pm$ 0.7	24.4 $\pm$ 2.3	27.6 $\pm$ 0.7	39.8 $\pm$ 1.1	34.4 $\pm$ 1.0
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	73.9 $\pm$ 0.3	47.1 $\pm$ 0.4	13.6 $\pm$ 0.7	20.5 $\pm$ 0.7	37.7 $\pm$ 4.0	37.1 $\pm$ 0.7	49.1 $\pm$ 0.5	39.9 $\pm$ 1.0
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	75.9 $\pm$ 0.4	49.9 $\pm$ 0.5	12.8 $\pm$ 1.0	20.0 $\pm$ 0.4	35.7 $\pm$ 1.1	37.7 $\pm$ 0.8	49.1 $\pm$ 0.5	40.1 $\pm$ 0.7

Table 2: k NN transfer for the five source regimes and methods of Table 1, classified by k NN on frozen features without a learned probe. Mean $\pm$ standard deviation over three seeds.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	7.0 $\pm$ 0.2	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.4
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	74.8 $\pm$ 0.4	41.0 $\pm$ 0.4	4.8 $\pm$ 0.5	10.0 $\pm$ 0.2	24.2 $\pm$ 0.6	16.2 $\pm$ 0.7	29.7 $\pm$ 0.3	28.7 $\pm$ 0.4
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	69.3 $\pm$ 0.5	39.5 $\pm$ 0.1	4.8 $\pm$ 0.6	8.9 $\pm$ 0.1	21.9 $\pm$ 0.1	15.5 $\pm$ 0.1	29.1 $\pm$ 0.7	27.0 $\pm$ 0.3
PASS-10k	SimCLR	67.2 $\pm$ 0.2	37.0 $\pm$ 0.3	5.4 $\pm$ 0.3	10.3 $\pm$ 0.5	30.4 $\pm$ 0.0	16.7 $\pm$ 0.5	38.3 $\pm$ 0.3	29.3 $\pm$ 0.3
	TS	65.2 $\pm$ 0.7	34.6 $\pm$ 0.3	4.4 $\pm$ 0.2	7.9 $\pm$ 0.2	24.5 $\pm$ 0.1	14.4 $\pm$ 0.4	39.3 $\pm$ 0.4	27.2 $\pm$ 0.3
	SDCLR	32.4 $\pm$ 1.5	11.5 $\pm$ 0.8	1.3 $\pm$ 0.2	2.0 $\pm$ 0.1	8.6 $\pm$ 0.7	5.2 $\pm$ 0.2	11.1 $\pm$ 0.8	10.3 $\pm$ 0.6
	BRIDGE (ours)	67.5 $\pm$ 0.3	36.9 $\pm$ 0.3	5.8 $\pm$ 0.5	10.7 $\pm$ 0.4	33.7 $\pm$ 0.2	17.8 $\pm$ 0.6	41.4 $\pm$ 0.8	30.6 $\pm$ 0.4
DiffusionDB-10k	SimCLR	67.3 $\pm$ 0.1	36.2 $\pm$ 0.4	4.3 $\pm$ 0.2	8.8 $\pm$ 0.1	26.8 $\pm$ 0.4	17.3 $\pm$ 0.1	36.7 $\pm$ 0.8	28.2 $\pm$ 0.3
	TS	64.7 $\pm$ 0.4	33.6 $\pm$ 0.3	3.1 $\pm$ 0.5	7.3 $\pm$ 0.3	23.2 $\pm$ 0.4	17.0 $\pm$ 0.0	37.8 $\pm$ 0.6	26.7 $\pm$ 0.4
	SDCLR	32.6 $\pm$ 1.2	10.0 $\pm$ 0.3	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.1 $\pm$ 0.7	4.8 $\pm$ 0.4	10.6 $\pm$ 0.5	9.7 $\pm$ 0.5
	BRIDGE (ours)	68.3 $\pm$ 0.2	37.4 $\pm$ 0.4	5.4 $\pm$ 0.3	9.8 $\pm$ 0.1	30.4 $\pm$ 0.2	18.6 $\pm$ 0.2	39.4 $\pm$ 1.0	29.9 $\pm$ 0.3

baselines, and BRIDGE trains five cycles of 100 epochs with $k = 100$, $B = 500$ and $G = 5$. DINO and MAE follow their published ViT recipes and are evaluated on ViT-S in paired experiments. Appendix D gives implementation details.

Compared methods and controls. The source-regime comparison covers source-matched SimCLR, TS, SDCLR and BRIDGE, with hybrids in the appendix. All ablations use ImageNet-100-LT source pretraining, the same downstream suite and three seeds, and vary one component at a time under the full five-cycle protocol. The generator ablation includes an oracle that returns the exact real images withheld when the long tail was constructed. This oracle is not a competing method, since it requires knowing which images are missing, information no label-free method has, and places true support at the true deficit. It is the ceiling for any repair operator at this budget, and a generator that reaches it recovers the full value of the withheld data without access to it. The single-stage controls of Section 6.4 branch from one source checkpoint, add the same number of images and receive the same optimizer updates.

Table 3: Paired five-cycle ViT-S experiments on ImageNet-100-LT across four SSL objectives, and end-to-end adaptation of the SimCLR ResNet-50 representations. Both arms of every pair branch from one source checkpoint and receive the same optimizer updates after the branch. Panel A reports per-task linear-probe accuracy with the seven-task linear-probe and k NN averages, mean $\pm$ standard deviation over three seeds, and per-task k NN accuracy is in Appendix Table 25. Panel B reports full fine-tuning with 1 and 10 percent of the ImageNet-100-LT labels and PASCAL VOC segmentation, three paired seeds.

A. Frozen ViT-S transfer under four objectives										
Objective	Method	Cars	Aircraft	Flowers	Pets	C10	C100	IN100-LT	Avg. linear	Avg. k NN
SimCLR	SSL baseline	7.2 $\pm$ 0.2	10.9 $\pm$ 0.4	32.9 $\pm$ 2.7	25.0 $\pm$ 1.2	60.5 $\pm$ 0.5	35.0 $\pm$ 0.2	42.0 $\pm$ 1.4	30.5	30.4
	with BRIDGE	8.2 $\pm$ 0.4	11.8 $\pm$ 0.4	36.6 $\pm$ 2.9	29.8 $\pm$ 3.0	60.9 $\pm$ 0.3	36.3 $\pm$ 0.1	43.9 $\pm$ 1.1	32.5	31.1
MoCo v3	SSL baseline	7.6 $\pm$ 0.2	11.1 $\pm$ 1.3	30.3 $\pm$ 1.4	29.3 $\pm$ 3.1	60.0 $\pm$ 0.4	34.5 $\pm$ 0.0	44.8 $\pm$ 1.2	31.1	33.0
	with BRIDGE	9.6 $\pm$ 0.3	13.4 $\pm$ 1.0	36.8 $\pm$ 6.1	31.2 $\pm$ 4.0	62.4 $\pm$ 0.5	36.3 $\pm$ 0.1	47.5 $\pm$ 0.9	33.9	32.8
DINO	SSL baseline	8.1 $\pm$ 0.2	11.4 $\pm$ 1.0	38.1 $\pm$ 5.2	27.5 $\pm$ 2.3	58.9 $\pm$ 0.9	33.6 $\pm$ 1.8	44.6 $\pm$ 1.5	31.8	30.5
	with BRIDGE	9.5 $\pm$ 0.9	13.3 $\pm$ 0.6	37.2 $\pm$ 2.6	34.6 $\pm$ 3.6	61.3 $\pm$ 0.1	35.2 $\pm$ 0.3	46.8 $\pm$ 0.8	34.0	32.0
MAE	SSL baseline	1.5 $\pm$ 0.2	2.4 $\pm$ 1.0	10.5 $\pm$ 5.9	10.1 $\pm$ 3.0	37.7 $\pm$ 3.7	13.2 $\pm$ 1.9	19.9 $\pm$ 2.1	13.6	9.6
	with BRIDGE	1.6 $\pm$ 0.4	3.2 $\pm$ 0.7	8.5 $\pm$ 2.3	8.6 $\pm$ 2.2	41.5 $\pm$ 2.2	15.0 $\pm$ 2.4	22.2 $\pm$ 2.4	14.4	11.6

B. End-to-end downstream adaptation, SimCLR ResNet-50				
Evaluation	Setting	SSL baseline	With BRIDGE	Paired difference
Full fine-tuning	IN100-LT, 1% labels	14.31 $\pm$ 0.33	16.29 $\pm$ 1.11	+1.97 $\pm$ 0.78
Full fine-tuning	IN100-LT, 10% labels	21.56 $\pm$ 0.75	24.72 $\pm$ 1.75	+3.17 $\pm$ 1.66
Segmentation	PASCAL VOC, mIoU	12.78 $\pm$ 0.79	15.20 $\pm$ 0.86	+2.42 $\pm$ 0.27

6 RESULTS

6.1 TRANSFER ACROSS FIVE SOURCE REGIMES

BRIDGE has the best seven-task average in every source regime under both protocols (Tables 1 and 2) and is the best method in 31 of the 35 linear-probe cells and 33 of the 35 k NN cells. The gain over source-matched SimCLR is largest where the source is smallest and most skewed, 10.6 and 5.1 linear-probe points under CIFAR-100-LT and CIFAR-10-LT, and smallest on the two web corpora, 3.1 on PASS and 2.4 on DiffusionDB, with the same ordering under k NN. TS leads only on the in-domain ImageNet-100-LT target under its own source and is below SimCLR on the fine-grained targets everywhere. SDCLR does not train well at this source size and schedule. Since the DiffusionDB source is itself synthetic and already carries a generator’s stylistic bias, its improvement cannot be attributed to harvesting natural web diversity.

6.2 BEYOND THE DEFAULT PROTOCOL

Table 3 tests whether the effect is specific to SimCLR on ResNet-50 or to frozen-feature evaluation. Four objectives run paired on ViT-S under one schedule and one repair budget, and BRIDGE raises the seven-task linear-probe average of every objective, by 2.0 points for SimCLR, 2.8 for MoCo v3, 2.2 for DINO and 0.7 for MAE. Of the 14 linear-probe and k NN measurements per objective, 12 improve for SimCLR, MoCo v3 and MAE and 13 for DINO, a two-sided sign test at $p < .013$ in every case, and the fine-grained targets gain most, up to 7.1 points on Pets for DINO. The same ResNet-50 representations also improve after end-to-end adaptation, by 2.0 and 3.2 points of full fine-tuning accuracy with 1 and 10 percent of the ImageNet-100-LT labels and by 2.4 mIoU on PASCAL VOC segmentation, each on three paired seeds.

6.3 ABLATIONS

Table 4 varies one component at a time under the full protocol.

Generator. Generated repair with Stable Diffusion 3 reaches 41.5 against 40.7 for the oracle that restores the withheld real images, so estimating where support is missing and filling it with variants of nearby images recovers the full value of knowing the missing images. FLUX at its original sampling settings reaches 35.0, and under matched settings FLUX.1-schnell reaches 19.68 ± 0.81 on the four-task diagnostic against 19.70 ± 0.35 for SD3 (Appendix Table 7), so the gap is a sampling-budget artifact.

Selection. The three mode windows with lower cutoffs at the 1st, 50th and 75th percentile reach 42.1 to 42.3 and do not differ detectably, while uniform acquisition reaches 35.5 ± 1.2 and the top

Table 4: Ablations on ImageNet-100-LT under the full five-cycle protocol, linear-probe accuracy, mean $\pm$ standard deviation over three seeds. Each block changes one component and holds the rest of the loop fixed. The oracle row restores the exact real images that were withheld when the long tail was constructed, so it knows where support is missing and is the ceiling for any repair operator at this budget. k NN counterparts are in Appendix Table 11.

Group	Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Generator	Stable Diffusion 3 (default)	75.0 $\pm$ 0.3	48.5 $\pm$ 0.2	16.3 $\pm$ 0.8	20.7 $\pm$ 0.4	35.7 $\pm$ 0.9	41.8 $\pm$ 0.8	52.4 $\pm$ 0.7	41.5 $\pm$ 0.6
	FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	10.3 $\pm$ 1.3	17.5 $\pm$ 0.9	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
	Oracle (withheld real images restored)	74.8 $\pm$ 0.4	47.8 $\pm$ 0.7	13.1 $\pm$ 0.1	20.5 $\pm$ 0.5	40.3 $\pm$ 6.1	38.1 $\pm$ 1.5	50.3 $\pm$ 0.4	40.7 $\pm$ 1.4
Selection	Mode window q1-99	75.6 $\pm$ 0.1	49.2 $\pm$ 0.2	16.9 $\pm$ 0.7	21.9 $\pm$ 0.4	34.6 $\pm$ 2.9	43.5 $\pm$ 0.7	52.8 $\pm$ 0.1	42.1 $\pm$ 0.7
	Mode window q50-99	75.5 $\pm$ 0.1	49.3 $\pm$ 0.4	15.2 $\pm$ 0.3	22.4 $\pm$ 0.6	35.5 $\pm$ 1.1	43.8 $\pm$ 3.1	53.4 $\pm$ 0.9	42.2 $\pm$ 0.9
	Mode window q75-99 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
	Top tail	70.8 $\pm$ 0.5	43.1 $\pm$ 1.0	10.0 $\pm$ 1.7	15.9 $\pm$ 1.3	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4	34.3 $\pm$ 1.2
	Uniform	71.5 $\pm$ 0.2	43.8 $\pm$ 0.7	9.3 $\pm$ 0.9	17.5 $\pm$ 1.2	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9	35.5 $\pm$ 1.2
Cycles	2 cycles	68.9 $\pm$ 0.6	41.6 $\pm$ 1.3	10.6 $\pm$ 1.2	16.4 $\pm$ 0.4	30.7 $\pm$ 4.5	33.2 $\pm$ 0.9	43.3 $\pm$ 0.6	35.0 $\pm$ 1.4
	5 cycles (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
	10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7
Architecture	ResNet-18	68.8 $\pm$ 0.1	41.7 $\pm$ 0.2	13.5 $\pm$ 0.3	19.7 $\pm$ 0.6	38.7 $\pm$ 3.4	38.4 $\pm$ 2.6	50.2 $\pm$ 0.2	38.7 $\pm$ 1.1
	ResNet-50 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
	ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	8.8 $\pm$ 0.5	12.2 $\pm$ 0.2	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
	ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	8.3 $\pm$ 0.5	14.3 $\pm$ 0.8	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

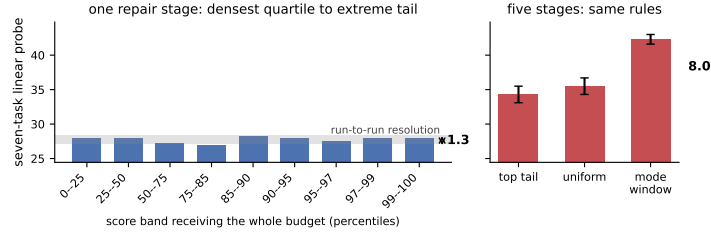


Figure 3: Left, nine bands of the score distribution each receive the whole budget after a single repair stage and lie within 1.3 points, the run-to-run resolution of the pipeline. Right, three acquisition rules under five stages at the same repair volume and optimizer budget separate by 8.0 points. The protocols differ only in how often the acquisition signal is recomputed.

tail 34.3 ± 1.2 , and every window is ahead of both after Holm correction over the eight comparisons at adjusted $p \leq .011$. The exact lower percentile is therefore not the operative choice, and trimming the extreme tail and diversifying within the remaining sparse band carries the 6.6 to 8.0 points. All five arms call the identical generator at identical settings, add the same number of images and receive the same updates, so an account in which the generator’s prior supplies the benefit predicts no spread across them, and the observed spread is 8.0 points.

Cycles and backbone. At fixed total repair volume and optimizer budget, five cycles improve the seven-task average over two by 7.3 points and remain 8.2 points ahead of ten, so the encoder needs enough training between repairs for its scores to reflect the inserted support. ResNet-50 leads the backbones at this source size, and Table 3 shows the repair effect on transformers. Appendix Table 20 varies the candidate-pool multiplier, upper cutoff, neighborhood size, distance metric and selection strategy one at a time, and none moves the seven-task average by more than the 1.3 points that separate cosine and normalized L_2 scoring, two rows that select the identical anchor set and so measure the run-to-run resolution.

6.4 WHERE THE ACQUISITION RULE ACQUIRES ITS EFFECT

Figure 3 spends one identical budget on each of nine bands of the score distribution and changes nothing else. After a single stage the densest quartile and the extreme tail are indistinguishable as places to spend the budget, as Proposition 5 predicts, and under five stages the same rules separate by 8.0 points. Appendix Table 6 places the acquisition rules of the closest prior methods under the same single-stage protocol. Uniform, DOPING-style extreme-tail, AIDE-style inverse cluster-frequency and TADA-style slowest-learned acquisition and the mode window all lie within 1.7 points, with the TADA-style rule the nominal leader at 20.00 ± 0.98 against 18.88 ± 0.58 for the mode window, inside the resolution of the protocol. A single batch of 2,500 well-placed images is worth no more than a badly placed one, and the value of a targeted rule accumulates only because every intervention

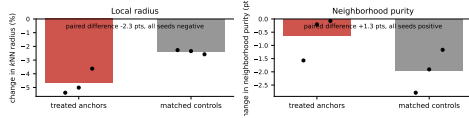


Figure 4: Fixed-encoder intervention test, one marker per seed. With the cycle-0 encoder frozen and every generated image inserted into the k NN index, treated anchors lose twice the relative radius of controls matched on initial radius, and their neighborhood purity is preserved while that of controls is diluted.

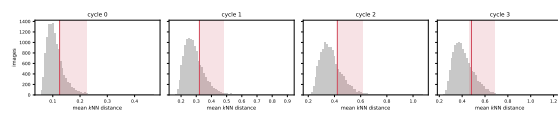


Figure 5: Score distribution over the four repair cycles of one ImageNet-100-LT run, 75th to 99th percentile band shaded and modal bin marked. The shape is stable while the scale grows, so the band is re-derived from the current representation at every cycle.

changes the representation that chooses the next, a regime for which TADA, DOPING and the cluster-occupancy engines, which decide once from a fixed model, are not defined. Conventional augmentation of the same anchors gives 18.5, so the operator does not separate arms at one stage either. The DiffAug adaptation, which learns its generator jointly with the encoder from the 10k source, reaches 11.5 and falls further when given the mode-window anchors, so a generator trained on the skewed source cannot repair it, and a pretrained generator held fixed is what makes targeted repair possible. Distilling the frozen generator’s features into the encoder without any generated image reaches 18.0 with source 1-NN accuracy of 26.3, below every image-adding arm, and adding BRIDGE on top returns both to 18.9 and 29.5.

6.5 REPRESENTATION GEOMETRY

Under the frozen cycle-0 encoder (Figure 4), inserting the generated images reduces the k NN radius of the treated anchors by 4.7 percent against 2.4 percent for controls matched one-to-one within class on initial radius, a paired difference of -2.3 points that is negative on every seed (Proposition 1, Appendix Table 8). Neighborhood label purity falls by 0.6 points at the anchors and by 2.0 at the controls, so the variants enter the anchors’ neighborhoods as same-class support. All 500 anchors chosen at cycle 0 have radii in the 75th to 99th percentile band, and Figure 5 shows why the band must be re-derived every cycle, since the score distribution keeps its shape while its scale grows. Global statistics of the learned space are objective-dependent (Appendix B).

7 LIMITATIONS

BRIDGE depends on informative encoder neighborhoods in the first cycle and on faithful local generation, and its frozen generator imports the coverage and biases of its own pretraining data. The score band, neighborhood size and budget are defaults that transferred across five sources without tuning, and no optimum has been shown. The fixed-encoder test verifies where support is inserted without establishing how every learned neighborhood evolves, later cycles may select generated anchors, and no generated-image filter is applied. In our unoptimized setting the loop raises wall-clock time over the no-repair run by 29 to 46 percent, almost all of it diffusion sampling at the default 20 steps without caching (Appendix E). A distilled few-step sampler, cached variants or generation on separate hardware would reduce that term directly, and since the generation budget is fixed while the pretraining cost grows with the source, the relative overhead falls with scale. Whether the fixed budget remains effective when the 10,000 added images are one percent of the source instead of 100 percent remains open.

8 CONCLUSION

Skewed-data SSL can be treated as a feedback problem in which the learner allocates a finite budget to change its own future source distribution, and under diminishing returns to local support and fidelity that decays in the extreme tail, the budget belongs in a sparse interior band and should be reallocated after each repair. BRIDGE implements that policy without labels, improves transfer over source-matched baselines on five sources and four objectives, matches an oracle that restores the withheld real images, and adds support where its encoder said it was missing. Where the budget is spent has no measurable effect for one intervention and an effect of 8.0 points once the representation that chooses the target has itself been repaired.

REPRODUCIBILITY STATEMENT

Section 4 and Algorithm 1 specify the loop, Section 5 and Appendix D give every operating parameter, schedule, generator setting and data split, and Appendix C contains complete proofs with their assumptions stated in Section 3. All results are three-seed means with standard deviations, paired comparisons share seeds and source checkpoints, and Appendix H reports every per-task number behind the main-text averages. An anonymous code release with configuration files for every experiment is linked in Appendix D.

ETHICS STATEMENT

The work uses public datasets under their release terms and a frozen pretrained image generator, releases no new generator, scraped collection or generated corpus, and involves no human subjects. Generative repair inherits the biases and memorization risks of the generator’s pretraining data, and a deployment would require a task-specific review of fairness, privacy and content safety, as Appendix G discusses.

AI USE STATEMENT

Generative AI tools were used to assist with editing the manuscript text, with writing and refactoring analysis and plotting code, and with checking bibliographic metadata against publisher records. They were not used to generate research ideas, experimental results or any number reported in this paper. All AI-assisted text and code were reviewed by the authors, who take responsibility for the final content of this work.

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A COMPARISON WITH RELATED ACQUISITION METHODS

Table 5 makes the comparison axes explicit. Classical oversampling and TADA receive labels or supervised learning signals. DOPING uses latent rarity, targets anomaly detection and does not close the loop through a changing SSL representation. GenRep and StableRep make synthetic images or latent neighborhoods the source of contrastive views without allocating a repair budget over an observed skewed corpus. DiffAug is the closest iterative SSL antecedent, and it jointly learns a generator for generic positive views without asking where a frozen generator should intervene. AIDE and industrial data-engine systems establish the model-and-data feedback pattern from supervised task failures with VLM and LLM reasoning, annotation and domain-specific verification. BRIDGE studies the label-free case in which local support in the current SSL representation is both the diagnostic and the acquisition signal.

Table 5: Closest methods and the dimensions relevant to the acquisition problem. Feedback means that the learner or task model changes the next acquisition decision, and the methods are cited in Section 2.

Method	Labels or task loss	Acquisition signal	Data action	Feedback	Primary goal
SMOTE	class labels	class minority	interpolate examples	no	supervised imbalance
DOPING	no labels	latent edge density	latent oversampling	no	anomaly detection
Concept pruning	no task labels	cluster complexity	remove examples	no	efficient pretraining
DA-Fusion	labels or few-shot set	class and exemplar	image-conditioned diffusion	no	supervised recognition
TADA	supervised loss dynamics	slow-learned examples	faithful diffusion variants	no	supervised recognition
GenRep and StableRep	prompts or generator latent	global sampling	synthetic SSL views	no	representation learning
DiffAug	no labels	learned positive-view objective	jointly learned diffusion views	yes	contrastive learning
AIDE	task labels and failures	VLM and LLM failure analysis	curate, generate, annotate	yes	driving detection
BRIDGE	no labels	local support in current SSL space	budgeted local variants	yes	source-support repair

Table 6: Acquisition and repair pipelines under one matched single-stage protocol on ImageNet-100-LT. Every arm branches from the same source checkpoint, receives the same optimizer budget and, where it generates, adds the same number of images. Transfer averages linear-probe accuracy on Cars, Aircraft, Flowers and Pets, and 1-NN accuracy is measured on the source set. Mean $\pm$ standard deviation over three seeds.

Family	Method	Four-task transfer	Source 1-NN
Acquisition rule	Uniform acquisition	18.33 ± 0.32	28.97 ± 2.11
	Top tail, DOPING-style	19.51 ± 0.22	28.63 ± 1.48
	Inverse cluster frequency, AIDE-style	18.66 ± 0.10	28.66 ± 2.42
	Slowest-learned examples, TADA-style	20.00 ± 0.98	28.78 ± 2.22
	Mode window, BRIDGE	18.88 ± 0.58	28.69 ± 1.82
Repair operator	Mode window, conventional augmentation	18.52 ± 0.65	28.91 ± 1.96
	DiffAug adaptation	11.50 ± 1.87	19.39 ± 2.78
	DiffAug adaptation with BRIDGE anchors	10.42 ± 2.17	20.28 ± 2.64
Generator prior	SD3 feature distillation, no images	18.02 ± 0.47	26.27 ± 2.29
	SD3 feature distillation with BRIDGE	18.87 ± 0.45	29.45 ± 2.26

Table 7: Generator sensitivity on the four-task diagnostic (Cars, Aircraft, Flowers, ImageNet-100-LT) after one repair stage from a common checkpoint, three paired seeds. The original FLUX setting of six steps and guidance 2.5 is the row that produced the 35.0 seven-task average in Table 4. Guidance under an empty prompt gives bit-identical images at 3.0, 5.0 and 7.5.

Generator	Setting	Four-task transfer
FLUX.1-dev	6 steps, guidance 2.5	18.23 ± 0.98
FLUX.1-dev	20 steps	18.78 ± 1.43
FLUX.1-schnell	20 steps	18.85 ± 1.30
FLUX.1-schnell	guidance 3	19.68 ± 0.81
SD3	6 steps	18.58 ± 1.14
SD3	20 steps (default)	19.70 ± 0.35
SD3	guidance 1.0 to 7.5	18.42 to 18.88
SD3	strength 0.3 to 0.9	18.60 to 19.63

Table 8: Fixed-encoder intervention test behind Figure 4. The cycle-0 encoder is frozen, every generated image of the run is inserted into the k NN index ($k = 20$), and the same original images are queried before and after. Anchors are the images selected at cycle 0, controls are matched one-to-one within class on initial radius and are never selected in any cycle. Radius change is relative to the initial radius, purity is the fraction of the 20 neighbors sharing the query label, and the last two columns are the anchor-minus-control differences.

Seed	Radius change (%)		Purity change (pts)		Difference	
	Anchors	Controls	Anchors	Controls	Radius	Purity
0	-5.38	-2.27	-1.57	-2.79	-3.11	+1.22
1	-5.01	-2.35	-0.20	-1.91	-2.66	+1.70
2	-3.62	-2.58	-0.07	-1.17	-1.05	+1.09
Mean	-4.67	-2.40	-0.61	-1.96	-2.27	+1.34

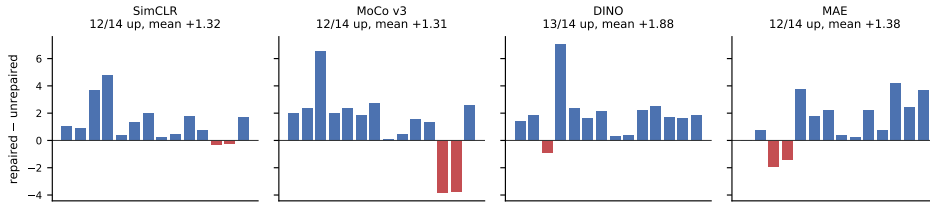


Figure 6: Repair against no repair for four objectives on ViT-S, one bar per downstream measurement (linear probe and k NN on each of seven datasets), from the paired runs of Table 3. Blue bars are improvements. The two losses under MoCo v3 are CIFAR-10 and CIFAR-100 under k NN.

B REPRESENTATION DIAGNOSTICS

The fixed-encoder intervention test freezes the cycle-0 encoder and queries the same original images before and after inserting all generated samples, so any change reflects the inserted points and never a change of the encoder. First-cycle anchors are matched one-to-one within class to original images with the closest initial k NN radius that are never selected in any cycle, and Table 8 reports the per-seed values behind Figure 4. The learned-space comparison retrain the encoder after the intervention and evaluates the same original-image panel. On that panel MoCo v3 on ViT-S gains 21.97 ± 1.58 in effective rank, 0.296 ± 0.019 in spectral entropy and 3.80 ± 0.80 points of 1-NN accuracy after repair, and the first two survive Holm correction, whereas SimCLR improves transfer while its effective rank falls and its local effect is mixed. We therefore treat the fixed-encoder result as evidence about where new support is inserted and the learned-space statistics as objective-dependent diagnostics.

Where the temperature-schedule hybrid helps. Combining BRIDGE with TS lowers the seven-task average in every source regime, and it wins 7 of the 70 individual cells (Appendix Tables 9 and 10). Six of those seven fall on CIFAR-10 and ImageNet-100-LT, the two long-tailed downstream targets, and the largest are +3.3 and +2.8 points on ImageNet-100-LT and CIFAR-10 under a PASS and a CIFAR-10-LT source. The fine-grained targets lose in almost every case. TS reweights the contrastive temperature toward rare samples, which sharpens the head-to-tail boundary that the long-tailed targets score, while BRIDGE adds support that broadens transfer. We read the hybrid as two interventions serving different targets and report the components separately.

C PROOFS FOR THE SUPPORT-ALLOCATION FRAMEWORK

Proposition 6 (Closed-form interior optimum). *Let $b(d) = b_0 d^\beta$ with $b_0 > 0$ and $\beta \in (0, 1]$, and let $\rho(d) = \rho_0 e^{-\gamma d}$ with $\gamma > 0$. Then g is strictly decreasing on $(0, \infty)$ for every $\lambda \geq 0$, so Proposition 3 applies whenever g changes sign on the interval. If $\lambda = 0$ the maximizer is $d^* = \beta/\gamma$, independent of b_0 and ρ_0 .*

Proof of Proposition 1. Sort the distances from z to points in Z and denote their k th order statistic by $r_k(z; Z)$. Adding A preserves every old candidate distance and may introduce smaller ones. The

k th order statistic can therefore only decrease. It decreases strictly when an added distance enters the first k positions and displaces a strictly larger distance. $\square$

Proof of Proposition 2. Let $x = n_j + m_j + \kappa$. Since $f(x) = x^{-\gamma}$ is positive, strictly decreasing and strictly convex for $x > 0$, the forward difference $f(x) - f(x+1)$ is positive and strictly decreases with x . Multiplication by $p_j a_j > 0$ preserves both properties. Equal $p_j a_j$ therefore leaves effective support as the sole determinant of the marginal ordering. $\square$

Proof of Proposition 3. Rewrite the utility as $u(d) = \rho(d)(b(d) + \lambda) - \lambda$. Its derivative is

$$u'(d) = \rho(d)(b(d) + \lambda) \left[\frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)} \right] = \rho(d)(b(d) + \lambda) g(d).$$

The factor preceding $g(d)$ is positive. Since g is strictly decreasing and changes sign across the interval, it has exactly one root d^* by continuity. Hence u increases before d^* and decreases after it, making d^* the unique global maximizer. $\square$

Proof of Proposition 6. With $b(d) = b_0 d^\beta$ and $\rho(d) = \rho_0 e^{-\gamma d}$ we have $\rho'(d)/\rho(d) = -\gamma$, so

$$g(d) = \frac{b_0 \beta d^{\beta-1}}{b_0 d^\beta + \lambda} - \gamma = \frac{\beta d^{\beta-1}}{d^\beta + c} - \gamma, \quad c = \lambda/b_0 \geq 0.$$

Write $h(d) = \beta d^{\beta-1}(d^\beta + c)^{-1}$. Differentiating,

$$h'(d) = \frac{\beta d^{\beta-2}[(\beta-1)(d^\beta + c) - \beta d^\beta]}{(d^\beta + c)^2} = \frac{\beta d^{\beta-2}[(\beta-1)c - d^\beta]}{(d^\beta + c)^2}.$$

For $\beta \in (0, 1]$ we have $(\beta-1)c \leq 0 < d^\beta$ for all $d > 0$, so the bracket is strictly negative and $h'(d) < 0$. Hence $g = h - \gamma$ is strictly decreasing for every $\lambda \geq 0$. When $\lambda = 0$ we have $c = 0$ and $g(d) = \beta/d - \gamma$, whose unique root is $d^* = \beta/\gamma$. Since g is positive below it and negative above it, d^* is the maximizer by the argument of Proposition 3. $\square$

Proof of Proposition 4. For region j define the marginal sequence $\delta_{j,t} = U_j(n_j+t) - U_j(n_j+t-1)$ for $t \geq 1$. Discrete concavity makes each sequence nonincreasing. Any feasible allocation of B repairs selects a prefix of length m_j from each sequence. Greedy allocation repeatedly chooses the largest available next element, and after B steps it has selected the B largest prefix-feasible marginals. If another allocation had larger utility, it would contain a selected marginal smaller than an available unselected marginal, and exchanging them would improve that allocation, which contradicts its optimality. A frozen allocation can fail this rule because it preserves the initial ordering after selected regions have moved to smaller marginals. $\square$

Proof of Proposition 5. Write $f(x) = x^{-\gamma}$. For $x > 0$ the mean value theorem gives $f(x) - f(x+1) = \gamma \xi^{-\gamma-1}$ for some $\xi \in (x, x+1)$, so

$$0 < \Delta_j(m_j) = p_j a_j [f(n_j + m_j + \kappa) - f(n_j + m_j + 1 + \kappa)] \leq p_j a_j \gamma (n_j + \kappa)^{-\gamma-1},$$

using $\xi > n_j + m_j + \kappa \geq n_j + \kappa$ and $m_j \geq 0$. Both $\mathcal{B}(\mathbf{m})$ and $\mathcal{B}(\mathbf{m}')$ are obtained from the common baseline $\mathcal{B}(\mathbf{0})$ by subtracting sums of at most B such marginals, since $\sum_j m_j = \sum_j m'_j = B$. Each sum lies in $[0, B\gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}]$, and the difference of two numbers drawn from an interval is bounded by its length. Monotonicity of $(n + \kappa)^{-\gamma-1}$ in n gives the $n_{\min}$ form. $\square$

D IMPLEMENTATION AND REPRODUCIBILITY

Training schedule. The encoder is warm-started from the previous stage at every cycle and is never retrained from scratch. Each stage uses a fresh optimizer and learning-rate schedule. All earlier real and generated images remain in the source set, so the setting differs from sequential-task learning in which previous data disappear. Five training stages contain four repairs, since generation runs only when another stage follows. The cycle ablation holds the total optimizer update budget and the total number of generated images fixed and compares fewer repairs with more learning between them against more frequent repairs with less.

Core configuration. The main experiments use ResNet-50 encoders, five cycles of 100 epochs, mean k NN distance with $k = 100$, 500 anchors per cycle and five variants per anchor. The optimizer is AdamW with weight decay 10^{-4} , and method-specific learning rates and temperatures follow the settings of each baseline. The paired ViT-S experiments use the reference recipe of each objective. DINO uses two global and eight local crops at 96×96 with a bottleneck projection head, and MAE runs five cycles of 80 epochs.

Scoring and selection. Scoring uses normalized ℓ_2 distance, and raw ℓ_2 , cosine and multiscale- k variants are reported as a robustness study in Table 20. The mode-window selector restricts the score distribution to the 75th to 99th percentile range, builds a histogram with the automatic bin heuristic, forms a mode-centered candidate pool of size $4B$ and applies greedy farthest-point sampling on normalized features to select the final B anchors.

Generators. The main method uses Stable Diffusion 3 Medium in image-to-image mode. With an empty prompt the guidance scale is inert, since at a fixed seed scales 3.0, 5.0 and 7.5 produce bit-identical images and scale 1.0 differs by a mean absolute pixel error of 0.0016 at 49.7 dB. The generation ablation replaces it with FLUX or with the oracle, which adds the withheld real images and runs no generator. In the PASS and DiffusionDB experiments the SD3 batch size is 12 to fit the 24-hour wall-time limit.

Single-stage controls. Every arm of Table 6 branches from one SimCLR ResNet-50 checkpoint per seed, adds 2,500 images where it generates, and receives the same number of optimizer updates. The TADA-style arm ranks source images by the decrease of their SSL loss over the first stage and selects the slowest-learned 500. The inverse cluster-frequency arm clusters the current embeddings and samples anchors with probability inverse to cluster occupancy. The conventional-augmentation arm applies the standard SimCLR augmentation pipeline five times to each mode-window anchor in place of the generator. The DiffAug arm trains a conditional diffusion generator jointly with the encoder on the source, following the published objective, and its BRIDGE variant conditions that generator on the mode-window anchors. The distillation arm adds a feature-matching loss between the encoder and frozen SD3 activations on the source images and adds no image.

Source data. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. The long-tailed ImageNet-100, CIFAR-10 and CIFAR-100 sources use their standard long-tailed configurations without the 10k restriction.

Seeds and reporting. All claims rest on three-seed averages reported as mean and standard deviation. Paired comparisons share seeds and, where stated, source checkpoints. Multiple comparisons within one table are corrected by the Holm procedure. Before publication we verify that no two conditions in a table share an encoder fingerprint.

Code availability. An anonymous code release is available at <https://anonymous.4open.science/status/FOMO-0488>.

E COMPUTE RESOURCES AND RUNTIME

Most experiments use one node with two NVIDIA A100 80GB GPUs under a 24-hour wall-time limit. The DINO arm fits on a single 48GB card at the batch size used. The 454 completed runs retained for this paper total approximately 5,471 wall-clock hours, of which the baselines and controls that appear only in the appendix account for 2,987. Runs on two-GPU nodes consume roughly twice that in GPU hours, while the paired ViT-S runs hold a single card each. These totals exclude failed runs, timeouts, cache migration and pilot jobs, so the project consumed more compute than the successful totals alone.

Representative mean wall-clock times over three seeds are as follows. On ImageNet-100-LT, SimCLR takes 16.9 hours and BRIDGE 21.8. On PASS, 10.2 and 14.9. On DiffusionDB, 11.6 and 16.4. The loop therefore raises wall-clock time by 29 to 46 percent over the corresponding no-repair run, of which the embedding pass is a small fraction and diffusion sampling the remainder. Four repairs add 10,000 images in total, which is 100 percent of a 10k source, 10 percent of a 100k source and 1

percent of a one-million-image source. This describes the relative cost under a fixed budget and does not establish that the fixed budget remains effective at those scales.

F ASSETS AND TERMS OF USE

ImageNet-100-LT is derived from ImageNet, whose access terms restrict the images to non-commercial research and educational use. PASS is released under Creative Commons Attribution 4.0 and DiffusionDB under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers and Oxford-IIIT Pets are used from their official academic releases under their original terms. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation under the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third-party datasets or generated image corpora.

G BROADER IMPACT

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful where balanced data are difficult to collect. The method also relies on large image generators and web-sourced datasets, which carry risks related to license compliance, privacy and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and release no new generators or scraped collections. Any deployment would require a task-specific review of fairness, privacy and content safety.

H FULL RESULTS

Table 9: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.8 $\pm$ 0.2	49.3 $\pm$ 0.2	16.4 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.5 $\pm$ 0.7
	BRIDGE+TS (ours)	74.2 $\pm$ 0.5	45.6 $\pm$ 0.8	9.4 $\pm$ 0.5	15.0 $\pm$ 0.2	25.5 $\pm$ 2.5	33.1 $\pm$ 2.8	48.3 $\pm$ 0.4	35.9 $\pm$ 1.1
	BRIDGE+SDCLR (ours)	43.8 $\pm$ 1.7	18.4 $\pm$ 0.7	2.0 $\pm$ 0.3	2.5 $\pm$ 0.8	10.0 $\pm$ 0.8	12.2 $\pm$ 3.1	21.8 $\pm$ 0.5	15.8 $\pm$ 1.1
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	78.1 $\pm$ 0.3	49.8 $\pm$ 0.4	9.2 $\pm$ 0.8	17.7 $\pm$ 1.0	18.2 $\pm$ 1.1	27.4 $\pm$ 2.7	39.0 $\pm$ 0.6	34.2 $\pm$ 1.0
	BRIDGE+TS (ours)	78.6 $\pm$ 0.3	46.1 $\pm$ 0.3	7.5 $\pm$ 0.9	11.1 $\pm$ 0.7	14.5 $\pm$ 0.6	25.1 $\pm$ 2.7	39.6 $\pm$ 0.8	31.8 $\pm$ 0.9
	BRIDGE+SDCLR (ours)	40.9 $\pm$ 0.7	17.9 $\pm$ 0.5	1.4 $\pm$ 0.1	2.2 $\pm$ 0.6	11.0 $\pm$ 2.4	10.0 $\pm$ 2.7	21.0 $\pm$ 0.6	14.9 $\pm$ 1.1
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	74.2 $\pm$ 0.5	48.1 $\pm$ 0.4	8.3 $\pm$ 1.0	18.2 $\pm$ 0.7	24.4 $\pm$ 2.3	27.6 $\pm$ 0.7	39.8 $\pm$ 1.1	34.4 $\pm$ 1.0
	BRIDGE+TS (ours)	73.7 $\pm$ 0.1	47.3 $\pm$ 0.5	7.9 $\pm$ 0.6	14.1 $\pm$ 0.5	18.2 $\pm$ 3.2	28.4 $\pm$ 0.4	41.3 $\pm$ 0.3	33.0 $\pm$ 0.8
	BRIDGE+SDCLR (ours)	42.0 $\pm$ 1.5	16.9 $\pm$ 0.9	1.3 $\pm$ 0.1	2.2 $\pm$ 0.6	10.6 $\pm$ 0.8	8.3 $\pm$ 0.8	20.2 $\pm$ 0.9	14.5 $\pm$ 0.8
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	73.9 $\pm$ 0.3	47.1 $\pm$ 0.4	13.6 $\pm$ 0.7	20.5 $\pm$ 0.7	37.7 $\pm$ 4.0	37.1 $\pm$ 0.7	49.1 $\pm$ 0.5	39.9 $\pm$ 1.0
	BRIDGE+TS (ours)	68.9 $\pm$ 0.6	40.6 $\pm$ 0.7	12.1 $\pm$ 0.4	14.1 $\pm$ 0.5	34.6 $\pm$ 2.9	31.6 $\pm$ 1.5	52.4 $\pm$ 0.6	36.4 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	44.0 $\pm$ 0.9	18.8 $\pm$ 0.9	1.4 $\pm$ 0.3	1.8 $\pm$ 0.9	12.6 $\pm$ 2.6	12.4 $\pm$ 1.3	21.2 $\pm$ 1.1	16.0 $\pm$ 1.1
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	75.9 $\pm$ 0.4	49.9 $\pm$ 0.5	12.8 $\pm$ 1.0	20.0 $\pm$ 0.4	35.7 $\pm$ 1.1	37.7 $\pm$ 0.8	49.1 $\pm$ 0.5	40.1 $\pm$ 0.7
	BRIDGE+TS (ours)	67.8 $\pm$ 0.6	39.1 $\pm$ 0.5	9.4 $\pm$ 1.2	12.4 $\pm$ 0.4	30.3 $\pm$ 3.0	33.7 $\pm$ 1.9	50.8 $\pm$ 0.6	34.8 $\pm$ 1.2
	BRIDGE+SDCLR (ours)	43.8 $\pm$ 0.9	18.5 $\pm$ 0.3	1.3 $\pm$ 0.4	2.5 $\pm$ 0.0	14.3 $\pm$ 1.6	11.4 $\pm$ 0.6	21.9 $\pm$ 0.6	16.2 $\pm$ 0.6

Table 11: Main k NN ablations on ImageNet-100-LT. Each block changes one component while holding the remaining pipeline fixed. The average is computed over all seven downstream k NN tasks.

Group	Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Generator	Stable Diffusion 3 (default)	68.6 $\pm$ 0.1	37.2 $\pm$ 0.2	6.4 $\pm$ 0.6	11.5 $\pm$ 0.2	35.2 $\pm$ 0.3	24.3 $\pm$ 0.6	46.8 $\pm$ 0.7	32.9 $\pm$ 0.4
	FLUX	65.8 $\pm$ 1.0	35.4 $\pm$ 0.8	5.7 $\pm$ 0.7	9.4 $\pm$ 0.9	27.7 $\pm$ 1.0	18.8 $\pm$ 1.3	38.7 $\pm$ 1.2	28.8 $\pm$ 1.0
	Oracle (withheld real images restored)	68.8 $\pm$ 0.2	37.7 $\pm$ 0.3	6.6 $\pm$ 0.5	11.4 $\pm$ 0.5	32.7 $\pm$ 0.1	23.6 $\pm$ 0.6	44.4 $\pm$ 0.4	32.2 $\pm$ 0.4
Selection	Mode window q1-99	69.2 $\pm$ 0.3	37.7 $\pm$ 0.2	6.6 $\pm$ 0.2	11.4 $\pm$ 0.2	35.9 $\pm$ 0.2	25.0 $\pm$ 0.5	48.3 $\pm$ 0.3	33.4 $\pm$ 0.3
	Mode window q50-99	68.7 $\pm$ 0.1	38.0 $\pm$ 0.2	7.3 $\pm$ 0.2	11.3 $\pm$ 0.2	36.5 $\pm$ 0.1	24.8 $\pm$ 0.8	48.5 $\pm$ 0.4	33.6 $\pm$ 0.3
	Mode window q75-99 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
	Top tail	65.7 $\pm$ 0.8	34.4 $\pm$ 0.2	4.1 $\pm$ 0.4	9.6 $\pm$ 0.2	23.9 $\pm$ 0.9	18.1 $\pm$ 0.8	35.4 $\pm$ 1.2	27.3 $\pm$ 0.6
	Uniform	66.1 $\pm$ 1.0	35.7 $\pm$ 0.3	4.9 $\pm$ 0.1	10.2 $\pm$ 0.7	25.3 $\pm$ 0.4	19.0 $\pm$ 0.8	37.4 $\pm$ 0.6	28.4 $\pm$ 0.6
Cycles	2 cycles	65.3 $\pm$ 0.4	34.2 $\pm$ 0.5	5.0 $\pm$ 0.7	10.0 $\pm$ 0.2	26.1 $\pm$ 0.8	18.7 $\pm$ 0.6	35.9 $\pm$ 0.3	27.9 $\pm$ 0.5
	5 cycles (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
	10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	10.4 $\pm$ 0.6	25.2 $\pm$ 0.8	18.6 $\pm$ 0.9	36.0 $\pm$ 0.7	28.1 $\pm$ 0.6
Architecture	ResNet-18	66.3 $\pm$ 0.2	35.7 $\pm$ 0.3	5.9 $\pm$ 0.2	11.1 $\pm$ 0.4	33.5 $\pm$ 0.1	23.3 $\pm$ 0.5	44.5 $\pm$ 0.7	31.5 $\pm$ 0.4
	ResNet-50 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
	ViT-S	67.3 $\pm$ 0.4	37.5 $\pm$ 0.3	5.3 $\pm$ 0.2	7.6 $\pm$ 0.1	36.2 $\pm$ 0.7	20.1 $\pm$ 1.0	42.1 $\pm$ 1.0	30.9 $\pm$ 0.5
	ViT-B	69.3 $\pm$ 0.8	40.1 $\pm$ 0.8	6.0 $\pm$ 0.2	8.3 $\pm$ 0.2	39.8 $\pm$ 0.8	21.9 $\pm$ 1.4	45.1 $\pm$ 0.7	32.9 $\pm$ 0.7

Table 10: Full k NN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	7.0 $\pm$ 0.2	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.4
	BRIDGE+TS (ours)	65.4 $\pm$ 0.2	33.6 $\pm$ 0.3	3.1 $\pm$ 0.1	7.1 $\pm$ 0.3	18.2 $\pm$ 0.2	15.6 $\pm$ 0.7	40.8 $\pm$ 0.5	26.3 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	36.3 $\pm$ 0.8	13.0 $\pm$ 0.5	1.1 $\pm$ 0.3	2.4 $\pm$ 0.2	9.7 $\pm$ 0.3	5.4 $\pm$ 0.4	13.9 $\pm$ 1.0	11.7 $\pm$ 0.5
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	74.8 $\pm$ 0.4	41.0 $\pm$ 0.4	4.8 $\pm$ 0.5	10.0 $\pm$ 0.2	24.2 $\pm$ 0.6	16.2 $\pm$ 0.7	29.7 $\pm$ 0.3	28.7 $\pm$ 0.4
	BRIDGE+TS (ours)	77.6 $\pm$ 0.7	35.3 $\pm$ 0.5	3.1 $\pm$ 0.3	6.6 $\pm$ 0.4	16.3 $\pm$ 0.6	12.5 $\pm$ 1.0	26.7 $\pm$ 0.1	25.5 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	34.4 $\pm$ 1.2	11.4 $\pm$ 0.6	1.2 $\pm$ 0.2	2.2 $\pm$ 0.2	8.8 $\pm$ 1.5	5.0 $\pm$ 0.2	11.7 $\pm$ 1.1	10.7 $\pm$ 0.7
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	69.3 $\pm$ 0.5	39.5 $\pm$ 0.1	4.8 $\pm$ 0.6	8.9 $\pm$ 0.1	21.9 $\pm$ 0.1	15.5 $\pm$ 0.1	29.1 $\pm$ 0.7	27.0 $\pm$ 0.3
	BRIDGE+TS (ours)	64.3 $\pm$ 0.4	36.6 $\pm$ 0.4	2.9 $\pm$ 0.3	6.3 $\pm$ 0.1	16.4 $\pm$ 0.5	12.3 $\pm$ 0.0	28.2 $\pm$ 0.6	23.9 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	31.9 $\pm$ 0.4	10.4 $\pm$ 0.5	0.8 $\pm$ 0.1	2.1 $\pm$ 0.5	7.5 $\pm$ 1.6	4.5 $\pm$ 0.1	9.1 $\pm$ 0.7	9.5 $\pm$ 0.6
PASS-10k	SimCLR	67.2 $\pm$ 0.2	37.0 $\pm$ 0.3	5.4 $\pm$ 0.3	10.3 $\pm$ 0.5	30.4 $\pm$ 0.0	16.7 $\pm$ 0.5	38.3 $\pm$ 0.3	29.3 $\pm$ 0.3
	TS	65.2 $\pm$ 0.7	34.6 $\pm$ 0.3	4.4 $\pm$ 0.2	7.9 $\pm$ 0.2	24.5 $\pm$ 0.1	14.4 $\pm$ 0.4	39.3 $\pm$ 0.4	27.2 $\pm$ 0.3
	SDCLR	32.4 $\pm$ 1.5	11.5 $\pm$ 0.8	1.3 $\pm$ 0.2	2.0 $\pm$ 0.1	8.6 $\pm$ 0.7	5.2 $\pm$ 0.2	11.1 $\pm$ 0.8	10.3 $\pm$ 0.6
	BRIDGE (ours)	67.5 $\pm$ 0.3	36.9 $\pm$ 0.3	5.8 $\pm$ 0.5	10.7 $\pm$ 0.4	33.7 $\pm$ 0.2	17.8 $\pm$ 0.6	41.4 $\pm$ 0.8	30.6 $\pm$ 0.4
	BRIDGE+TS (ours)	58.8 $\pm$ 0.5	27.9 $\pm$ 0.5	3.0 $\pm$ 0.2	6.6 $\pm$ 0.1	23.0 $\pm$ 0.3	13.4 $\pm$ 0.4	38.8 $\pm$ 0.8	24.5 $\pm$ 0.4
	BRIDGE+SDCLR (ours)	37.1 $\pm$ 2.0	13.6 $\pm$ 1.0	1.6 $\pm$ 0.2	2.4 $\pm$ 0.2	10.1 $\pm$ 0.5	6.3 $\pm$ 0.3	13.1 $\pm$ 0.6	12.0 $\pm$ 0.7
DiffusionDB-10k	SimCLR	67.3 $\pm$ 0.1	36.2 $\pm$ 0.4	4.3 $\pm$ 0.2	8.8 $\pm$ 0.1	26.8 $\pm$ 0.4	17.3 $\pm$ 0.1	36.7 $\pm$ 0.8	28.2 $\pm$ 0.3
	TS	64.7 $\pm$ 0.4	33.6 $\pm$ 0.3	3.1 $\pm$ 0.5	7.3 $\pm$ 0.3	23.2 $\pm$ 0.4	17.0 $\pm$ 0.0	37.8 $\pm$ 0.6	26.7 $\pm$ 0.4
	SDCLR	32.6 $\pm$ 1.2	10.0 $\pm$ 0.3	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.1 $\pm$ 0.7	4.8 $\pm$ 0.4	10.6 $\pm$ 0.5	9.7 $\pm$ 0.5
	BRIDGE (ours)	68.3 $\pm$ 0.2	37.4 $\pm$ 0.4	5.4 $\pm$ 0.3	9.8 $\pm$ 0.1	30.4 $\pm$ 0.2	18.6 $\pm$ 0.2	39.4 $\pm$ 1.0	29.9 $\pm$ 0.3
	BRIDGE+TS (ours)	57.5 $\pm$ 0.8	26.6 $\pm$ 0.3	3.0 $\pm$ 0.4	6.5 $\pm$ 0.1	21.7 $\pm$ 0.8	15.5 $\pm$ 0.4	37.0 $\pm$ 0.4	24.0 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	35.2 $\pm$ 0.7	11.1 $\pm$ 0.3	1.0 $\pm$ 0.1	2.2 $\pm$ 0.1	8.1 $\pm$ 1.0	5.0 $\pm$ 0.3	11.3 $\pm$ 0.7	10.6 $\pm$ 0.5

Table 12: Augmentation mechanism ablation with full linear-probe results. The oracle row is not a competing generator. It restores the exact real images withheld when the long tail was built, so it knows the identity and location of the missing support, which the method only estimates, and it is the ceiling for any repair operator at this budget.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	75.0 $\pm$ 0.3	48.5 $\pm$ 0.2	16.3 $\pm$ 0.8	20.7 $\pm$ 0.4	35.7 $\pm$ 0.9	41.8 $\pm$ 0.8	52.4 $\pm$ 0.7	41.5 $\pm$ 0.6
FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	10.3 $\pm$ 1.3	17.5 $\pm$ 0.9	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
Oracle (withheld real images restored)	74.8 $\pm$ 0.4	47.8 $\pm$ 0.7	13.1 $\pm$ 0.1	20.5 $\pm$ 0.5	40.3 $\pm$ 6.1	38.1 $\pm$ 1.5	50.3 $\pm$ 0.4	40.7 $\pm$ 1.4

Table 13: Augmentation mechanism ablation with full k NN results. The oracle row is not a competing generator. It restores the exact real images withheld when the long tail was built, so it knows the identity and location of the missing support, which the method only estimates, and it is the ceiling for any repair operator at this budget.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	68.6 $\pm$ 0.1	37.2 $\pm$ 0.2	6.4 $\pm$ 0.6	11.5 $\pm$ 0.2	35.2 $\pm$ 0.3	24.3 $\pm$ 0.6	46.8 $\pm$ 0.7	32.9 $\pm$ 0.4
FLUX	65.8 $\pm$ 1.0	35.4 $\pm$ 0.8	5.7 $\pm$ 0.7	9.4 $\pm$ 0.9	27.7 $\pm$ 1.0	18.8 $\pm$ 1.3	38.7 $\pm$ 1.2	28.8 $\pm$ 1.0
Oracle (withheld real images restored)	68.8 $\pm$ 0.2	37.7 $\pm$ 0.3	6.6 $\pm$ 0.5	11.4 $\pm$ 0.5	32.7 $\pm$ 0.1	23.6 $\pm$ 0.6	44.4 $\pm$ 0.4	32.2 $\pm$ 0.4

Table 14: Selection rule ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	75.6 $\pm$ 0.1	49.2 $\pm$ 0.2	16.9 $\pm$ 0.7	21.9 $\pm$ 0.4	34.6 $\pm$ 2.9	43.5 $\pm$ 0.7	52.8 $\pm$ 0.1	42.1 $\pm$ 0.7
Mode-window q50-99 + diversity	75.5 $\pm$ 0.1	49.3 $\pm$ 0.4	15.2 $\pm$ 0.3	22.4 $\pm$ 0.6	35.5 $\pm$ 1.1	43.8 $\pm$ 3.1	53.4 $\pm$ 0.9	42.2 $\pm$ 0.9
Mode-window q75-99 + diversity (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
Top-tail	70.8 $\pm$ 0.5	43.1 $\pm$ 1.0	10.0 $\pm$ 1.7	15.9 $\pm$ 1.3	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4	34.3 $\pm$ 1.2
Uniform	71.5 $\pm$ 0.2	43.8 $\pm$ 0.7	9.3 $\pm$ 0.9	17.5 $\pm$ 1.2	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9	35.5 $\pm$ 1.2

Table 15: Selection rule ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	69.2 $\pm$ 0.3	37.7 $\pm$ 0.2	6.6 $\pm$ 0.2	11.4 $\pm$ 0.2	35.9 $\pm$ 0.2	25.0 $\pm$ 0.5	48.3 $\pm$ 0.3	33.4 $\pm$ 0.3
Mode-window q50-99 + diversity	68.7 $\pm$ 0.1	38.0 $\pm$ 0.2	7.3 $\pm$ 0.2	11.3 $\pm$ 0.2	36.5 $\pm$ 0.1	24.8 $\pm$ 0.8	48.5 $\pm$ 0.4	33.6 $\pm$ 0.3
Mode-window q75-99 + diversity (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
Top-tail	65.7 $\pm$ 0.8	34.4 $\pm$ 0.2	4.1 $\pm$ 0.4	9.6 $\pm$ 0.2	23.9 $\pm$ 0.9	18.1 $\pm$ 0.8	35.4 $\pm$ 1.2	27.3 $\pm$ 0.6
Uniform	66.1 $\pm$ 1.0	35.7 $\pm$ 0.3	4.9 $\pm$ 0.1	10.2 $\pm$ 0.7	25.3 $\pm$ 0.4	19.0 $\pm$ 0.8	37.4 $\pm$ 0.6	28.4 $\pm$ 0.6

Table 16: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	68.9 $\pm$ 0.6	41.6 $\pm$ 1.3	10.6 $\pm$ 1.2	16.4 $\pm$ 0.4	30.7 $\pm$ 4.5	33.2 $\pm$ 0.9	43.3 $\pm$ 0.6	35.0 $\pm$ 1.4
5 cycles (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7

Table 17: Cycle count ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.3 $\pm$ 0.4	34.2 $\pm$ 0.5	5.0 $\pm$ 0.7	10.0 $\pm$ 0.2	26.1 $\pm$ 0.8	18.7 $\pm$ 0.6	35.9 $\pm$ 0.3	27.9 $\pm$ 0.5
5 cycles (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	10.4 $\pm$ 0.6	25.2 $\pm$ 0.8	18.6 $\pm$ 0.9	36.0 $\pm$ 0.7	28.1 $\pm$ 0.6

Table 18: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	68.8 $\pm$ 0.1	41.7 $\pm$ 0.2	13.5 $\pm$ 0.3	19.7 $\pm$ 0.6	38.7 $\pm$ 3.4	38.4 $\pm$ 2.6	50.2 $\pm$ 0.2	38.7 $\pm$ 1.1
ResNet-50 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	8.8 $\pm$ 0.5	12.2 $\pm$ 0.2	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	8.3 $\pm$ 0.5	14.3 $\pm$ 0.8	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

Table 19: Backbone architecture ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	66.3 $\pm$ 0.2	35.7 $\pm$ 0.3	5.9 $\pm$ 0.2	11.1 $\pm$ 0.4	33.5 $\pm$ 0.1	23.3 $\pm$ 0.5	44.5 $\pm$ 0.7	31.5 $\pm$ 0.4
ResNet-50 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
ViT-S	67.3 $\pm$ 0.4	37.5 $\pm$ 0.3	5.3 $\pm$ 0.2	7.6 $\pm$ 0.1	36.2 $\pm$ 0.7	20.1 $\pm$ 1.0	42.1 $\pm$ 1.0	30.9 $\pm$ 0.5
ViT-B	69.3 $\pm$ 0.8	40.1 $\pm$ 0.8	6.0 $\pm$ 0.2	8.3 $\pm$ 0.2	39.8 $\pm$ 0.8	21.9 $\pm$ 1.4	45.1 $\pm$ 0.7	32.9 $\pm$ 0.7

Table 20: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean $\pm$ standard deviation over three seeds. Cosine scoring rescales the normalized L_2 score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized L_2 row measures the end-to-end run-to-run spread of the pipeline. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Candidate pool $\alpha = 1$	6.46 $\pm$ 0.57	11.37 $\pm$ 1.59	23.53 $\pm$ 3.00	21.92 $\pm$ 2.00	61.94 $\pm$ 0.94	32.73 $\pm$ 0.21	33.92 $\pm$ 1.68
Candidate pool $\alpha = 2$	6.77 $\pm$ 0.63	11.59 $\pm$ 0.40	21.79 $\pm$ 3.22	23.19 $\pm$ 3.93	62.90 $\pm$ 1.06	34.48 $\pm$ 0.11	35.69 $\pm$ 0.87
Candidate pool $\alpha = 4$ (default)	6.31 $\pm$ 0.10	11.17 $\pm$ 3.78	23.53 $\pm$ 3.27	24.18 $\pm$ 2.16	62.11 $\pm$ 0.19	33.39 $\pm$ 0.64	34.18 $\pm$ 0.47
Candidate pool $\alpha = 8$	6.41 $\pm$ 0.28	11.16 $\pm$ 0.78	23.09 $\pm$ 4.35	24.82 $\pm$ 3.29	62.60 $\pm$ 1.79	33.15 $\pm$ 1.01	35.40 $\pm$ 0.87
Upper cutoff $q = 0.95$	6.79 $\pm$ 0.65	11.44 $\pm$ 0.85	19.83 $\pm$ 2.47	25.27 $\pm$ 2.32	62.57 $\pm$ 0.87	33.83 $\pm$ 0.53	35.13 $\pm$ 1.03
Upper cutoff $q = 0.99$ (default)	6.59 $\pm$ 0.17	11.76 $\pm$ 1.54	20.26 $\pm$ 1.73	24.00 $\pm$ 2.20	63.11 $\pm$ 0.75	33.43 $\pm$ 1.02	35.85 $\pm$ 1.15
Upper cutoff $q = 0.995$	6.78 $\pm$ 0.66	10.40 $\pm$ 2.49	22.88 $\pm$ 4.58	22.55 $\pm$ 3.09	62.76 $\pm$ 0.50	33.87 $\pm$ 0.38	35.32 $\pm$ 0.64
Upper cutoff $q = 1.0$	6.31 $\pm$ 0.45	10.94 $\pm$ 1.34	22.44 $\pm$ 1.89	23.37 $\pm$ 4.00	62.85 $\pm$ 1.04	33.91 $\pm$ 0.87	35.51 $\pm$ 1.59
Neighbourhood $k = 10$	6.26 $\pm$ 0.29	10.44 $\pm$ 1.43	22.22 $\pm$ 0.65	24.00 $\pm$ 1.23	62.66 $\pm$ 0.60	33.44 $\pm$ 0.58	35.14 $\pm$ 1.21
Neighbourhood $k = 25$	6.44 $\pm$ 0.76	11.13 $\pm$ 1.19	19.39 $\pm$ 3.60	23.37 $\pm$ 1.18	63.18 $\pm$ 0.56	33.40 $\pm$ 1.21	33.49 $\pm$ 0.43
Neighbourhood $k = 50$	6.34 $\pm$ 0.56	10.01 $\pm$ 1.32	20.48 $\pm$ 1.00	22.64 $\pm$ 3.24	61.28 $\pm$ 2.19	32.64 $\pm$ 0.29	34.23 $\pm$ 0.77
Neighbourhood $k = 200$	6.50 $\pm$ 0.46	10.58 $\pm$ 2.83	20.26 $\pm$ 1.13	22.46 $\pm$ 1.23	61.24 $\pm$ 1.08	32.87 $\pm$ 0.76	34.42 $\pm$ 1.23
Distance: normalized L_2 (default)	6.59 $\pm$ 0.17	11.79 $\pm$ 1.34	20.48 $\pm$ 2.64	24.46 $\pm$ 2.68	61.69 $\pm$ 0.67	33.88 $\pm$ 1.05	35.24 $\pm$ 1.10
Distance: cosine (null control)	6.55 $\pm$ 0.15	10.50 $\pm$ 1.10	20.04 $\pm$ 2.64	23.91 $\pm$ 1.70	62.22 $\pm$ 1.29	33.23 $\pm$ 0.81	34.98 $\pm$ 0.51
Distance: raw L_2	6.92 $\pm$ 0.78	10.45 $\pm$ 3.00	18.95 $\pm$ 5.23	25.72 $\pm$ 2.31	62.81 $\pm$ 0.80	33.20 $\pm$ 0.89	35.08 $\pm$ 2.48
Strategy: mode window (default)	6.55 $\pm$ 0.15	10.18 $\pm$ 1.30	22.44 $\pm$ 4.59	23.28 $\pm$ 2.04	60.95 $\pm$ 3.01	33.05 $\pm$ 0.21	34.53 $\pm$ 0.92
Strategy: sparse band with FPS	6.60 $\pm$ 0.62	11.21 $\pm$ 1.26	21.13 $\pm$ 2.64	22.46 $\pm$ 0.16	63.34 $\pm$ 0.27	34.29 $\pm$ 0.30	34.30 $\pm$ 1.26
Strategy: sparse band, random	6.46 $\pm$ 0.48	11.06 $\pm$ 0.70	21.35 $\pm$ 4.82	22.92 $\pm$ 0.87	62.26 $\pm$ 0.89	32.33 $\pm$ 0.96	34.46 $\pm$ 1.08
Strategy: inverse-cluster	6.44 $\pm$ 0.75	10.66 $\pm$ 1.06	19.17 $\pm$ 2.47	21.65 $\pm$ 2.28	62.70 $\pm$ 0.45	33.50 $\pm$ 0.95	34.69 $\pm$ 1.10
Strategy: densest window	6.64 $\pm$ 0.46	11.90 $\pm$ 0.49	20.04 $\pm$ 1.00	24.28 $\pm$ 1.64	63.50 $\pm$ 0.60	33.80 $\pm$ 0.93	35.20 $\pm$ 0.80
Strategy: top tail, no diversity	6.39 $\pm$ 0.43	10.48 $\pm$ 1.03	19.83 $\pm$ 2.00	23.82 $\pm$ 1.50	62.86 $\pm$ 0.92	33.14 $\pm$ 1.02	34.66 $\pm$ 0.41

Table 21: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean $\pm$ standard deviation over three seeds. Cosine scoring rescales the normalized L_2 score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized L_2 row measures the end-to-end run-to-run spread of the pipeline. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Candidate pool $\alpha = 1$	5.02 $\pm$ 0.14	8.62 $\pm$ 0.26	19.22 $\pm$ 0.71	14.92 $\pm$ 0.99	62.40 $\pm$ 0.12	31.47 $\pm$ 0.25	33.47 $\pm$ 1.05
Candidate pool $\alpha = 2$	5.08 $\pm$ 0.15	8.36 $\pm$ 0.22	19.07 $\pm$ 0.42	15.17 $\pm$ 0.27	62.81 $\pm$ 0.58	31.72 $\pm$ 0.27	33.19 $\pm$ 0.61
Candidate pool $\alpha = 4$ (default)	4.92 $\pm$ 0.15	8.48 $\pm$ 0.72	19.04 $\pm$ 0.03	14.60 $\pm$ 0.38	62.35 $\pm$ 0.54	31.84 $\pm$ 0.37	32.59 $\pm$ 0.49
Candidate pool $\alpha = 8$	4.85 $\pm$ 0.17	8.53 $\pm$ 0.41	19.17 $\pm$ 0.90	14.82 $\pm$ 0.62	62.15 $\pm$ 0.07	31.56 $\pm$ 0.23	33.58 $\pm$ 1.03
Upper cutoff $q = 0.95$	4.99 $\pm$ 0.11	8.57 $\pm$ 0.51	19.09 $\pm$ 0.51	14.68 $\pm$ 0.28	62.84 $\pm$ 0.45	31.88 $\pm$ 0.04	32.56 $\pm$ 0.45
Upper cutoff $q = 0.99$ (default)	4.95 $\pm$ 0.17	8.53 $\pm$ 0.69	19.40 $\pm$ 0.45	14.85 $\pm$ 0.49	62.29 $\pm$ 0.50	31.87 $\pm$ 0.16	32.44 $\pm$ 0.48
Upper cutoff $q = 0.995$	5.01 $\pm$ 0.07	8.34 $\pm$ 0.74	19.58 $\pm$ 0.83	14.87 $\pm$ 0.48	63.05 $\pm$ 0.47	32.06 $\pm$ 0.11	33.01 $\pm$ 0.34
Upper cutoff $q = 1.0$	4.99 $\pm$ 0.08	8.15 $\pm$ 0.61	18.52 $\pm$ 0.98	14.45 $\pm$ 0.19	62.84 $\pm$ 0.32	31.69 $\pm$ 0.30	32.99 $\pm$ 0.83
Neighbourhood $k = 10$	5.04 $\pm$ 0.09	8.99 $\pm$ 1.21	18.86 $\pm$ 0.35	14.90 $\pm$ 0.57	62.73 $\pm$ 0.39	31.69 $\pm$ 0.17	33.14 $\pm$ 0.40
Neighbourhood $k = 25$	4.97 $\pm$ 0.14	8.59 $\pm$ 0.21	19.05 $\pm$ 0.29	15.04 $\pm$ 0.85	62.68 $\pm$ 0.66	31.82 $\pm$ 0.16	33.56 $\pm$ 0.97
Neighbourhood $k = 50$	5.06 $\pm$ 0.17	8.62 $\pm$ 0.77	19.26 $\pm$ 0.67	14.92 $\pm$ 0.03	62.63 $\pm$ 0.40	31.71 $\pm$ 0.59	31.88 $\pm$ 0.57
Neighbourhood $k = 200$	5.17 $\pm$ 0.15	8.65 $\pm$ 0.59	19.22 $\pm$ 1.02	15.17 $\pm$ 0.10	62.80 $\pm$ 0.31	31.90 $\pm$ 0.13	33.91 $\pm$ 0.54
Distance: normalized L_2 (default)	4.97 $\pm$ 0.18	8.55 $\pm$ 0.67	19.42 $\pm$ 0.49	14.85 $\pm$ 0.50	62.27 $\pm$ 0.52	31.86 $\pm$ 0.16	32.60 $\pm$ 0.96
Distance: cosine (null control)	4.90 $\pm$ 0.13	8.52 $\pm$ 0.69	19.35 $\pm$ 0.54	14.75 $\pm$ 0.44	62.31 $\pm$ 0.53	31.78 $\pm$ 0.28	32.44 $\pm$ 0.83
Distance: raw L_2	4.85 $\pm$ 0.10	7.83 $\pm$ 0.39	19.13 $\pm$ 0.88	14.63 $\pm$ 0.34	62.81 $\pm$ 0.52	31.71 $\pm$ 0.04	32.28 $\pm$ 0.65
Strategy: mode window (default)	4.90 $\pm$ 0.12	8.52 $\pm$ 0.69	19.34 $\pm$ 0.54	14.75 $\pm$ 0.45	62.31 $\pm$ 0.52	31.78 $\pm$ 0.28	32.73 $\pm$ 0.75
Strategy: sparse band with FPS	5.19 $\pm$ 0.04	8.12 $\pm$ 0.39	19.53 $\pm$ 0.49	14.95 $\pm$ 0.75	62.43 $\pm$ 0.17	31.62 $\pm$ 0.32	33.38 $\pm$ 1.46
Strategy: sparse band, random	5.00 $\pm$ 0.13	8.70 $\pm$ 0.06	19.36 $\pm$ 0.88	14.95 $\pm$ 0.61	62.63 $\pm$ 0.19	31.64 $\pm$ 0.62	33.04 $\pm$ 0.82
Strategy: inverse-cluster	5.05 $\pm$ 0.02	8.83 $\pm$ 1.05	19.40 $\pm$ 0.81	15.06 $\pm$ 0.46	62.60 $\pm$ 0.50	31.99 $\pm$ 0.37	33.75 $\pm$ 0.44
Strategy: densest window	4.88 $\pm$ 0.10	8.44 $\pm$ 0.54	19.34 $\pm$ 0.59	14.95 $\pm$ 0.53	62.46 $\pm$ 0.12	31.62 $\pm$ 0.30	33.08 $\pm$ 0.64
Strategy: top tail, no diversity	4.90 $\pm$ 0.12	8.38 $\pm$ 0.29	19.11 $\pm$ 0.64	15.22 $\pm$ 0.03	62.05 $\pm$ 0.77	31.77 $\pm$ 0.13	32.41 $\pm$ 0.25

Table 22: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the k NN score distribution, holding every other part of the loop fixed. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
q0–25 (densest)	6.73 $\pm$ 0.36	9.63 $\pm$ 0.26	24.84 $\pm$ 5.10	23.37 $\pm$ 2.72	62.97 $\pm$ 0.82	33.96 $\pm$ 0.49	34.44 $\pm$ 0.62
q25–50	6.80 $\pm$ 0.39	10.28 $\pm$ 1.37	23.75 $\pm$ 2.10	22.74 $\pm$ 1.57	62.76 $\pm$ 0.64	33.20 $\pm$ 1.12	35.61 $\pm$ 0.23
q50–75	6.79 $\pm$ 0.63	10.24 $\pm$ 1.94	20.04 $\pm$ 4.64	23.55 $\pm$ 2.58	62.24 $\pm$ 0.37	32.80 $\pm$ 1.66	34.51 $\pm$ 1.27
q75–85	6.12 $\pm$ 0.33	9.74 $\pm$ 1.00	22.88 $\pm$ 4.58	23.46 $\pm$ 1.64	60.97 $\pm$ 1.05	31.90 $\pm$ 0.36	33.55 $\pm$ 0.46
q85–90	6.62 $\pm$ 0.39	9.95 $\pm$ 0.73	23.75 $\pm$ 3.09	25.45 $\pm$ 1.77	63.38 $\pm$ 0.78	33.47 $\pm$ 0.93	35.24 $\pm$ 0.69
q90–95	6.55 $\pm$ 0.55	10.46 $\pm$ 1.87	22.66 $\pm$ 3.94	25.27 $\pm$ 0.72	62.30 $\pm$ 1.15	33.49 $\pm$ 0.87	34.40 $\pm$ 0.62
q95–97	6.58 $\pm$ 0.20	9.55 $\pm$ 1.62	23.09 $\pm$ 1.00	23.55 $\pm$ 0.68	61.77 $\pm$ 2.13	32.75 $\pm$ 2.27	34.99 $\pm$ 1.19
q97–99	6.80 $\pm$ 0.17	11.38 $\pm$ 2.82	20.26 $\pm$ 3.98	26.27 $\pm$ 3.24	62.73 $\pm$ 1.49	33.13 $\pm$ 1.97	35.31 $\pm$ 1.55
q99–100 (extreme tail)	6.55 $\pm$ 0.28	10.78 $\pm$ 1.23	21.79 $\pm$ 0.75	23.82 $\pm$ 1.37	63.54 $\pm$ 0.92	33.65 $\pm$ 0.55	35.57 $\pm$ 0.91

Table 23: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the k NN score distribution, holding every other part of the loop fixed. Mean $\pm$ standard deviation over three seeds. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
q0–25 (densest)	4.97 $\pm$ 0.07	8.33 $\pm$ 0.72	20.00 $\pm$ 0.65	14.48 $\pm$ 0.27	62.04 $\pm$ 0.84	31.45 $\pm$ 0.05	31.88 $\pm$ 0.91
q25–50	5.16 $\pm$ 0.14	8.52 $\pm$ 0.37	19.58 $\pm$ 0.49	14.82 $\pm$ 0.73	62.95 $\pm$ 0.79	31.54 $\pm$ 0.09	32.88 $\pm$ 0.78
q50–75	5.06 $\pm$ 0.08	8.51 $\pm$ 0.44	19.63 $\pm$ 0.75	14.46 $\pm$ 0.81	62.41 $\pm$ 0.61	31.47 $\pm$ 0.56	32.61 $\pm$ 0.55
q75–85	4.97 $\pm$ 0.08	8.78 $\pm$ 0.45	19.54 $\pm$ 0.14	15.39 $\pm$ 0.20	62.52 $\pm$ 0.56	31.31 $\pm$ 0.43	32.23 $\pm$ 1.17
q85–90	5.04 $\pm$ 0.09	9.04 $\pm$ 0.48	19.61 $\pm$ 0.46	14.75 $\pm$ 0.65	62.57 $\pm$ 0.27	31.74 $\pm$ 0.34	32.82 $\pm$ 0.96
q90–95	4.92 $\pm$ 0.21	8.67 $\pm$ 0.75	19.60 $\pm$ 0.53	14.96 $\pm$ 0.46	62.40 $\pm$ 0.59	31.56 $\pm$ 0.54	32.48 $\pm$ 0.56
q95–97	4.88 $\pm$ 0.35	8.51 $\pm$ 1.09	19.40 $\pm$ 0.72	14.97 $\pm$ 0.49	62.32 $\pm$ 0.94	31.55 $\pm$ 0.43	31.63 $\pm$ 0.32
q97–99	5.02 $\pm$ 0.08	8.59 $\pm$ 0.43	19.63 $\pm$ 0.67	14.89 $\pm$ 0.56	62.64 $\pm$ 0.56	31.75 $\pm$ 0.61	32.59 $\pm$ 0.96
q99–100 (extreme tail)	5.17 $\pm$ 0.16	8.16 $\pm$ 0.23	19.10 $\pm$ 0.60	15.00 $\pm$ 0.53	62.81 $\pm$ 0.26	32.12 $\pm$ 0.43	32.15 $\pm$ 1.40

I ALGORITHM

Algorithm 1 BRIDGE training loop with mode-window selection

- 1: Initialize source set $D^{(0)}$, encoder $f_{\theta^{(0)}}$, frozen generator g_ϕ , stage lengths $\{E_c\}_{c=0}^{C-1}$
- 2: **for** $c = 0, \dots, C - 1$ **do**
- 3: Continue training $f_{\theta^{(c)}}$ on $D^{(c)}$ for E_c epochs with a fresh optimizer schedule
- 4: **if** $c < C - 1$ **then**
- 5: Compute normalized embeddings $Z^{(c)} = \{f_{\theta^{(c)}}(x_i) \mid x_i \in D^{(c)}\}$
- 6: Compute mean k NN scores $\{d_i^{(c)}\}$ from $Z^{(c)}$
- 7: Restrict to the 75th to 99th percentile band and locate the modal bin center $m^{(c)}$
- 8: Form the candidate pool $C^{(c)}$ of size αB whose scores are closest to $m^{(c)}$
- 9: Select $S^{(c)} \subseteq C^{(c)}$ with $|S^{(c)}| = B$ by greedy farthest-point sampling in embedding space
- 10: Form $\tilde{D}^{(c)} = \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$ and set $D^{(c+1)} = D^{(c)} \cup \tilde{D}^{(c)}$
- 11: **end if**
- 12: **end for**
- 13: Return the final encoder $f_{\theta^{(C-1)}}$

Table 24: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	7.15 $\pm$ 0.19	10.92 $\pm$ 0.40	32.90 $\pm$ 2.72	25.00 $\pm$ 1.25	60.46 $\pm$ 0.46	34.96 $\pm$ 0.20	41.95 $\pm$ 1.43
SimCLR, with BRIDGE	8.18 $\pm$ 0.42	11.82 $\pm$ 0.43	36.60 $\pm$ 2.85	29.80 $\pm$ 2.98	60.85 $\pm$ 0.35	36.33 $\pm$ 0.14	43.93 $\pm$ 1.09
MoCo v3, SSL baseline	7.57 $\pm$ 0.23	11.05 $\pm$ 1.31	30.28 $\pm$ 1.36	29.26 $\pm$ 3.13	59.96 $\pm$ 0.39	34.47 $\pm$ 0.01	44.79 $\pm$ 1.17
MoCo v3, with BRIDGE	9.55 $\pm$ 0.34	13.42 $\pm$ 0.98	36.82 $\pm$ 6.07	31.25 $\pm$ 4.00	62.36 $\pm$ 0.46	36.30 $\pm$ 0.13	47.55 $\pm$ 0.86
DINO, SSL baseline	8.10 $\pm$ 0.25	11.40 $\pm$ 1.03	38.13 $\pm$ 5.24	27.54 $\pm$ 2.34	58.93 $\pm$ 0.94	33.56 $\pm$ 1.75	44.63 $\pm$ 1.47
DINO, with BRIDGE	9.49 $\pm$ 0.88	13.28 $\pm$ 0.58	37.25 $\pm$ 2.61	34.60 $\pm$ 3.62	61.30 $\pm$ 0.07	35.23 $\pm$ 0.31	46.78 $\pm$ 0.77
MAE, SSL baseline	1.54 $\pm$ 0.19	2.44 $\pm$ 1.00	10.46 $\pm$ 5.88	10.05 $\pm$ 3.03	37.69 $\pm$ 3.73	13.20 $\pm$ 1.94	19.93 $\pm$ 2.13
MAE, with BRIDGE	1.56 $\pm$ 0.43	3.18 $\pm$ 0.70	8.50 $\pm$ 2.26	8.61 $\pm$ 2.20	41.46 $\pm$ 2.20	15.00 $\pm$ 2.40	22.19 $\pm$ 2.35

Table 25: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean $\pm$ standard deviation over three seeds. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	6.26 $\pm$ 0.17	7.09 $\pm$ 0.17	36.53 $\pm$ 0.89	19.16 $\pm$ 0.64	66.05 $\pm$ 0.48	36.72 $\pm$ 0.36	41.21 $\pm$ 0.72
SimCLR, with BRIDGE	6.52 $\pm$ 0.22	7.56 $\pm$ 0.23	38.29 $\pm$ 0.83	19.93 $\pm$ 0.53	65.70 $\pm$ 0.85	36.46 $\pm$ 0.33	42.91 $\pm$ 0.78
MoCo v3, SSL baseline	6.65 $\pm$ 0.38	7.71 $\pm$ 0.16	38.91 $\pm$ 1.03	23.56 $\pm$ 1.12	69.29 $\pm$ 0.25	39.98 $\pm$ 0.44	45.14 $\pm$ 0.46
MoCo v3, with BRIDGE	6.74 $\pm$ 0.26	8.21 $\pm$ 0.18	40.51 $\pm$ 1.45	24.91 $\pm$ 0.45	65.41 $\pm$ 0.21	36.24 $\pm$ 0.46	47.71 $\pm$ 1.53
DINO, SSL baseline	6.05 $\pm$ 0.27	7.73 $\pm$ 0.52	38.52 $\pm$ 1.19	19.26 $\pm$ 0.16	62.69 $\pm$ 2.42	33.98 $\pm$ 2.80	45.28 $\pm$ 0.98
DINO, with BRIDGE	6.39 $\pm$ 0.22	8.16 $\pm$ 0.26	40.72 $\pm$ 0.31	21.79 $\pm$ 1.15	64.38 $\pm$ 3.51	35.59 $\pm$ 2.51	47.14 $\pm$ 0.29
MAE, SSL baseline	1.00 $\pm$ 0.14	2.24 $\pm$ 0.29	5.30 $\pm$ 0.52	4.58 $\pm$ 0.17	31.34 $\pm$ 1.65	10.96 $\pm$ 0.44	11.75 $\pm$ 0.38
MAE, with BRIDGE	1.38 $\pm$ 0.37	2.50 $\pm$ 0.70	7.54 $\pm$ 2.13	5.35 $\pm$ 0.76	35.58 $\pm$ 7.14	13.44 $\pm$ 3.46	15.46 $\pm$ 2.68